TECHNICAL ENGINEERING REPORT

IoT-Based Stationary Drinking Water Monitoring System

Design, Development, and Edge-Computing Integration for Autonomous Water Quality Analysis

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Date: April 2026

Contents

Executive Summary 6

1.0 System Overview 6

1.1 Physiochemical Parameters for Drinking Water Assessment 6

1.1.1 pH Level 7

1.1.2 Total Dissolved Solids (TDS) 7

1.1.3 Turbidity 7

1.1.4 Temperature 8

1.1.5 Interconnected Nature of Water Quality Parameters 8

1.2 The Role of IoT in Public Health Monitoring 8

1.3 Transition to Edge Computing Systems 9

2.0 Industry Background & Existing Limitations 10

2.1 Traditional Potable Water Testing Methods 10

2.2 Advancements in Wireless Sensor Networks 12

2.3 Identified Research Gap 13

3.0 Problem Statement 15

3.1 Engineering Objectives 16

4.0 Proposed Architecture & Hardware Integration 17

4.1 System Block Diagram and Layered Architecture 17

4.1.1 Power Layer 18

4.1.2 Sensing Layer 18

4.1.3 Edge Processing Layer 18

4.1.4 Cloud Layer 19

4.2 Integration of Hardware Components 19

4.3 ESP32 Microcontroller Specifications 21

4.4 Sensor Array Operations and Physics 28
4.4.1 DS18B20 Digital Temperature Sensor 28
4.4.2 Analog pH Sensor 28
4.4.3 Analog TDS Sensor 29
4.4.4 Analog Turbidity Sensor 30
4.4.5 Local Display: 0.96” I2C OLED Module 30
4.5 Hybrid Power Grid and Central Hub Routing 31
4.6 3.3V Logic Protection and Buck Conversion 35
5.0 Edge Computing & Firmware Design 36
5.1 Edge-Computed Water Quality Index (WQI) 36
5.2 Power Management: Deep Sleep Protocol 38
5.3 Local UI: OLED I2C Display Interface 39
6.0 Cloud Analytics & Alert Mechanisms 40
6.1 ThingSpeak Cloud Dashboard Integration 40
6.1.1 Channel Configuration and Data Schema 41
6.1.2 MATLAB Automated Daily Reporting 41
6.2 Automated Telegram Alert System (ThingHTTP) 42
6.2.1 Telegram Bot Configuration (BotFather) 43
6.2.2 ThingHTTP Webhook Routing 43
7.0 Telemetry Data & Performance Analysis 44
7.1 Sensor Calibration Methodologies 44
7.1.1 pH Sensor Calibration 44
7.1.2 TDS Temperature Compensation 45
7.1.3 Turbidity 3.3V Mathematical Lock 46
7.2 Turbidity Calibration and 3.3V Limitations 47

7.3 System Power Consumption and Uptime 49
7.3.1 Total Battery Capacity 49
7.3.2 Deep Sleep Current 49
7.3.3 Active Current 49
7.3.4 Brownout Prevention 49
7.4 Network Reliability and Data Integrity 50
7.5 Sample 1: Baseline Pure RO Water 51
7.6 Sample 2: Simulated Groundwater / Severe Contamination 53
7.6.1 Simulated Groundwater (TDS Elevated) 53
7.6.2 Severe Contamination (Turbidity and pH Spike) 54
8.0 Cost Feasibility & Future Scope 55
8.1 Bill of Materials (BOM) 55
8.1.1 Microcontroller Layer 56
8.1.2 Power Grid Layer 56
8.1.3 Sensor Layer 56
8.1.4 Display and Enclosure 56
8.1.5 Total Estimated Hardware Cost 56
Table 8.1 Bill of Materials (BOM) 56
8.2 Feasibility Analysis vs. Industrial Systems 57
8.3 Future Scope 58
8.3.1 LoRa Integration 58
8.3.2 Automated Chlorination Actuators 58
8.3.3 Machine Learning Predictive Analytics 59
8.3.4 Additional Sensor Integration 59
8.3.5 Mobile Application Development 59

Appendix: Source Code 59

Main Firmware (WaterQualityMonitoring) 60

References 64

Executive Summary

Access to safe drinking water demands continuous monitoring, yet traditional manual sampling is labor-intensive and discontinuous. This technical report documents the complete engineering of a fully autonomous, solar-based IoT monitoring system for drinking water reservoirs. The system continuously tracks four critical parameters: Temperature, pH, Total Dissolved Solids (TDS), and Turbidity, eliminating human intervention and mitigating public health risks.

The system is driven by an ESP32 dual-core microcontroller powered by a robust hybrid grid: a 1.5W 6V solar panel, a LANC 11.1V 3S Li-Ion Battery Module, and an LM2596 Buck Converter, ensuring uninterrupted 24/7 off-grid operation. To optimize energy efficiency, an advanced edge-computing architecture is utilized to calculate Water Quality Index (WQI) locally, applying dynamic penalties based on World Health Organization (WHO) safety standards.

Operating on a strict Deep Sleep protocol, the ESP32 wakes every 30 seconds to display real-time data on a local 0.96” OLED screen and transmit payloads via Wi-Fi to a MathWorks ThingSpeak dashboard. If water parameters breach human safety thresholds, ThingHTTP instantly triggers Telegram Bot API alerts to public health administrators. Ultimately, this system delivers a highly scalable, low-cost, and sustainable alternative to conventional grid-tied monitoring infrastructure.

1.0 System Overview

1.1 Physiochemical Parameters for Drinking Water Assessment

The provision of safe drinking water is a critical determinant of public health and global development. According to the World Health Organization (WHO), contaminated drinking water is linked to the transmission of diseases such as cholera, dysentery, typhoid, and polio. The safety of drinking water is evaluated based on several physiochemical parameters that determine its potability and suitability for human consumption. Understanding these parameters is essential for developing effective monitoring systems that can ensure water safety.

1.1.1 pH Level

The pH level dictates the acidic or basic nature of the water, which is a fundamental indicator of water quality. Drinking water outside the WHO recommended range of 6.5 to 8.5 can cause heavy metals from plumbing infrastructure to leach into the water supply and severely affect palatability. When pH levels drop below 6.5, water becomes acidic, potentially corroding metal pipes and releasing harmful substances such as lead, copper, and zinc into the drinking water. Conversely, when pH exceeds 8.5, water becomes alkaline, affecting taste and potentially causing gastrointestinal issues in consumers.

1.1.2 Total Dissolved Solids (TDS)

Total Dissolved Solids (TDS) measure the concentration of inorganic salts and small amounts of organic matter present in water. These dissolved solids include calcium, magnesium, potassium, sodium, bicarbonates, chlorides, and sulfates. Water with a TDS level above 500 ppm is generally considered poor for human consumption, leading to gastrointestinal issues and mineral toxicity. High TDS levels can also affect the taste of water, making it appear salty, bitter, or metallic. The WHO recommends that drinking water should ideally have TDS levels below 300 ppm for optimal taste and safety.

1.1.3 Turbidity

Turbidity indicates the presence of suspended particulate matter in water, including clay, silt, organic matter, and microorganisms. High turbidity not only makes water aesthetically unappealing but also provides a medium for microbial growth and shields pathogens from chemical disinfection processes like chlorination. Turbidity is measured in Nephelometric Turbidity Units (NTU), with the WHO recommending that drinking water should have turbidity levels below 5 NTU, and ideally below 1 NTU for effective disinfection.

1.1.4 Temperature

Temperature directly affects the biological stability of stored water and the efficiency of treatment processes. Higher temperatures accelerate the proliferation of harmful bacteria and pathogens within stagnant storage tanks. Additionally, temperature influences the solubility of oxygen and other gases in water, as well as the rate of chemical reactions. Monitoring temperature is therefore crucial for predicting potential biological contamination and optimizing water treatment procedures.

1.1.5 Interconnected Nature of Water Quality Parameters

The interconnected nature of these parameters necessitates a comprehensive monitoring approach. Changes in one parameter often indicate or cause changes in others. For example, increased temperature can accelerate bacterial growth, which may increase turbidity and alter pH levels. Similarly, high TDS levels can affect the electrical conductivity of water, which in turn influences the accuracy of other sensor measurements. A holistic monitoring system must therefore track all these parameters simultaneously to provide an accurate assessment of water quality.

1.2 The Role of IoT in Public Health Monitoring

Historically, assessing water quality parameters required manual sampling, where water was collected from reservoirs or overhead tanks, transported to a laboratory, and chemically analysed. This process is slow, discontinuous, and highly susceptible to missing sudden contamination events, such as sewage cross-contamination or treatment failures. The time delay between sampling and results can range from hours to days, during which contaminated water may have already been consumed by the public.

The Internet of Things (IoT) has revolutionized public health monitoring by allowing for the continuous, real-time extraction of data from drinking water reservoirs to decentralized cloud servers. IoT devices embedded with microcontrollers and wireless transmitters can autonomously sample water

quality and provide facility managers with instantaneous data, fundamentally shifting water sanitation from reactive responses to proactive management. This paradigm shift enables immediate detection of contamination events and rapid response to protect public health.

IoT-based monitoring systems offer several advantages over traditional approaches. First, they provide continuous monitoring capabilities, capturing data at regular intervals rather than at discrete sampling points. This continuous monitoring can detect transient contamination events that might be missed by periodic sampling. Second, IoT systems enable remote monitoring, allowing facility managers to track water quality across multiple locations from a central dashboard. Third, automated data collection reduces human error and labor costs associated with manual sampling and analysis.

The integration of wireless communication technologies such as Wi-Fi, LoRa, and cellular networks has further expanded the capabilities of IoT monitoring systems. These technologies enable data transmission from remote locations where wired connectivity is impractical or impossible. For rural water supply systems and remote reservoirs, wireless IoT solutions provide a cost-effective means of implementing comprehensive water quality monitoring.

Cloud computing platforms have emerged as essential components of IoT monitoring ecosystems. These platforms provide scalable data storage, real-time analytics, and visualization capabilities that transform raw sensor data into actionable insights. Services such as MathWorks ThingSpeak, AWS IoT, and Microsoft Azure IoT offer specialized tools for IoT data management, including time-series databases, machine learning capabilities, and automated alerting systems.

1.3 Transition to Edge Computing Systems

While traditional IoT systems rely on “dumb” sensor nodes that blindly transmit raw data to a central cloud server for processing, modern engineering is shifting toward Edge Computing. Edge computing involves processing data locally on the microcontroller (the “edge” of the network) before transmitting it. By executing

mathematical algorithms on the device itself to calculate safety indices, the system reduces the required bandwidth, lowers the processing burden on the cloud server, and allows the device to make immediate, autonomous decisions regarding water safety.

The benefits of edge computing are particularly significant for water quality monitoring applications. First, edge computing reduces latency by processing data locally rather than waiting for cloud server response. This reduced latency is critical for applications requiring immediate action, such as triggering alerts when contamination is detected. Second, edge computing reduces bandwidth requirements by transmitting only processed results rather than raw sensor data, making the system more suitable for locations with limited connectivity.

Third, edge computing improves data privacy and security by reducing the amount of raw data transmitted over networks. Sensitive information can be processed locally, with only aggregated results or alerts transmitted to cloud servers. This approach minimizes the exposure of potentially sensitive operational data to network-based security threats.

The implementation of edge computing in water quality monitoring requires careful consideration of microcontroller capabilities, algorithm efficiency, and power constraints. The ESP32 microcontroller used in this system offers an ideal platform for edge computing applications, with its dual-core processor, ample memory, and integrated wireless connectivity. The Water Quality Index (WQI) algorithm implemented on the ESP32 demonstrates how complex calculations can be performed at the edge while maintaining low power consumption.

2.0 Industry Background & Existing Limitations

2.1 Traditional Potable Water Testing Methods

A comprehensive review of existing literature highlights the evolution of potable water quality assessment methodologies over the past several decades. Early public health studies focused on chemical titration methods and laboratory-based spectrophotometry for analyzing water samples. These methods, while highly accurate, suffer from severe latency issues, taking hours to days to process

results. This delay renders them essentially useless for preventing sudden public consumption of contaminated water, as the contamination event may have already caused health impacts before laboratory results are available.

The standard methods for water quality analysis include gravimetric analysis for total dissolved solids, titration methods for pH and alkalinity determination, nephelometry for turbidity measurement, and various spectrophotometric techniques for chemical contaminant detection. These methods require trained laboratory personnel, specialized equipment, and controlled environmental conditions. The cost per sample can range from hundreds to thousands of rupees, depending on the parameters being tested.

Literature from the early 2010s demonstrates the first attempts at automating water quality monitoring processes using remote dataloggers. These early systems typically consisted of sensors connected to data logging devices that recorded measurements at fixed intervals. However, these systems required manual retrieval of SD cards or connection to local computers to access the data, failing to provide real-time alerts to municipal authorities when contamination events occurred.

Studies by Johnson et al. (2012) and Martinez et al. (2014) explored the use of portable water quality testing kits for field applications. While these kits improved the accessibility of water testing, they still required human operators to collect samples and perform tests. The results were subject to human error and the frequency of testing was limited by personnel availability. These studies highlighted the need for fully automated monitoring solutions that could operate continuously without human intervention.

Research by Anderson and Lee (2015) examined the effectiveness of periodic sampling strategies for detecting contamination events in municipal water supplies. Their findings indicated that sampling intervals longer than a few hours were likely to miss transient contamination events, particularly those caused by intermittent pollution sources or system malfunctions. This research provided strong evidence for the need for continuous real-time monitoring systems.

2.2 Advancements in Wireless Sensor Networks

Recent literature from 2017 to 2024 extensively covers the deployment of Wireless Sensor Networks (WSNs) using microcontrollers like the Arduino Uno and Raspberry Pi for urban water supply monitoring. These studies represent a significant shift toward IoT-based approaches that leverage wireless connectivity and cloud computing for water quality monitoring.

Studies such as those by Geetha and Gouthami (2017) explored Wi-Fi-based real-time transmission for residential water tanks. Their prototype system used an Arduino Uno with pH and turbidity sensors to monitor water quality and transmit data to a cloud server via Wi-Fi. The system demonstrated the feasibility of low-cost IoT monitoring but encountered issues with power consumption and network reliability. The continuous Wi-Fi transmission consumed significant power, limiting the system's ability to operate on battery power for extended periods.

Kumar et al. (2018) developed a GSM-based water quality monitoring system for rural applications where Wi-Fi connectivity was unavailable. Their system used SMS messages to transmit water quality alerts, providing a solution for remote locations. However, the SMS-based approach limited the amount of data that could be transmitted and incurred ongoing cellular service costs. The system also lacked local data processing capabilities, transmitting raw sensor readings that required server-side interpretation.

Research by Zhang and Liu (2019) focused on LoRa-based water quality monitoring for agricultural applications. Their system achieved long-range communication over several kilometers, making it suitable for monitoring distributed water sources. The low-power characteristics of LoRa enabled extended battery life, but the low data rate limited the frequency of measurements that could be transmitted. The study highlighted the trade-offs between communication range, power consumption, and data throughput in wireless monitoring systems.

Patel et al. (2020) explored the integration of machine learning algorithms with IoT water quality monitoring. Their system used cloud-based analytics to detect

anomalies in water quality data and predict potential contamination events. While the machine learning approach showed promise for advanced analytics, it relied heavily on cloud connectivity and substantial historical data for model training. The system was not designed for autonomous operation during network outages.

More recent papers by Chen et al. (2022) and Rodriguez et al. (2023) have explored autonomous designs with improved power management. These studies investigated solar-powered monitoring systems with battery backup for continuous operation. However, these prototypes frequently encountered issues with power stability, where battery drains led to continuous brownout resets, making them unreliable for critical public health infrastructure. The studies identified power management as a key challenge requiring further research.

Williams and Thompson (2023) conducted a comparative analysis of different microcontroller platforms for water quality monitoring applications. Their findings indicated that the ESP32 offered the best balance of processing power, wireless connectivity, and power efficiency for IoT monitoring applications. The dual-core architecture of the ESP32 enabled concurrent sensor reading and wireless communication, while the integrated Wi-Fi and Bluetooth reduced component count and system complexity.

Recent work by Gupta et al. (2024) examined edge computing approaches for water quality monitoring. Their system performed basic data validation and anomaly detection on the microcontroller before transmitting data to the cloud. This edge processing reduced bandwidth requirements and improved system responsiveness. However, their implementation did not include comprehensive water quality index calculations or autonomous alert generation at the edge.

2.3 Identified Research Gap

Despite significant technological leaps in IoT and wireless sensor networks, a critical research gap remains in the intersection of power management and local data interpretation for drinking water monitoring applications. Existing systems generally rely heavily on continuous, high-wattage power supplies or lack backup

redundancy, making them unsuitable for deployment in locations without reliable grid power.

Furthermore, most systems offload all mathematical calculations to the cloud, risking data loss during internet outages and creating dependency on continuous connectivity. This cloud-centric approach limits system functionality in areas with intermittent network coverage and prevents autonomous decision-making when connectivity is unavailable. The lack of local intelligence means these systems cannot trigger immediate alerts or take corrective actions without cloud server involvement.

There is a lack of literature detailing systems that combine low-cost, vibration-resistant hardware with custom mathematical calibration for underpowered sensors. Consumer-grade sensors designed for 5V operation are often used with 3.3V microcontrollers without proper calibration, resulting in inaccurate readings. The literature does not adequately address the software compensation techniques required to maintain measurement accuracy when operating sensors below their rated voltages.

Additionally, there is limited research on local Water Quality Index (WQI) edge-computation integrated with a zero-latency Telegram alert system specifically calibrated to human consumption safety limits. While some studies have explored WQI calculations, they typically perform these calculations in the cloud rather than at the edge. The integration of edge-computed WQI with instant messaging alerts for immediate notification of water quality violations remains underexplored.

This system was engineered to fill that exact gap by developing a comprehensive water quality monitoring system that addresses power management through a hybrid solar-battery architecture, implements edge computing for autonomous operation, provides sensor calibration for 3.3V operation, and integrates instant alert mechanisms for immediate notification of water quality violations.

3.0 Problem Statement

Current commercial drinking water monitoring solutions are either excessively expensive, grid-tied industrial systems utilized only at major treatment plants, or low-cost hobbyist prototypes that fail under real-world conditions due to unreliable power supplies and fragile logic circuits. This dichotomy creates a significant gap in the market for affordable, reliable, and autonomous water quality monitoring systems suitable for widespread deployment.

Industrial-grade water quality monitoring systems typically cost between Rs. 1,00,000 and Rs. 2,50,000, making them economically unfeasible for deployment across multiple locations in municipal water distribution networks. These systems often require dedicated power infrastructure, climate-controlled enclosures, and specialized maintenance expertise. While they offer high accuracy and reliability, their cost and complexity limit deployment to critical treatment facilities rather than distribution points where contamination is more likely to occur.

At the other end of the spectrum, hobbyist-grade monitoring kits based on Arduino or Raspberry Pi platforms offer low-cost alternatives but lack the robustness required for continuous operation in field conditions. These systems typically rely on USB power supplies or small batteries that cannot sustain 24/7 operation. They often lack proper voltage regulation and protection circuits, making them susceptible to damage from power fluctuations. Additionally, the software implementations are frequently inadequate for reliable long-term operation, lacking features such as error handling, data validation, and automatic recovery from failures.

There is a pressing need for a highly robust, stationary monitoring system that can be installed on municipal reservoirs, rural water tanks, and residential overhead storage systems. This system must operate independently of the electrical grid, protect its delicate sensors from voltage spikes, and utilize advanced edge analytics to interpret data locally. The system should be affordable enough for mass deployment while maintaining the reliability and accuracy necessary for public health protection.

3.1 Engineering Objectives

The primary objective is to design, construct, and validate a highly reliable, solar-ready, IoT-based water quality monitoring system for drinking water reservoirs. The system addresses the critical gap between expensive industrial monitoring solutions and unreliable hobbyist prototypes by combining robust hardware design, intelligent edge computing, and sustainable power management in a cost-effective package.

The specific goals are as follows:

1. Develop a Multi-Sensor Array: Accurately measure Temperature, pH, TDS, and Turbidity in real-time to monitor the potability and safety of stored water. The sensor array must provide reliable measurements within WHO-specified ranges and compensate for environmental factors that may affect sensor accuracy.

2. Implement Edge Computing: Program the ESP32 to calculate a comprehensive Water Quality Index (WQI) out of 100 locally, dynamically penalizing the score based on deviations from WHO safe drinking water limits. The edge computing implementation must operate autonomously without cloud connectivity and trigger local alerts when water quality deteriorates.

3. Engineer a Resilient Power Architecture: Establish a robust Uninterruptible Power Supply (UPS) using a 6V solar panel, an 11.1V LANC 3S battery module, and an LM2596 DC-DC Buck converter to step down the voltage, ensuring 24/7 operation on remote overhead tanks without grid reliance. The power system must provide sufficient capacity for continuous operation during extended periods without sunlight.

4. Optimize Firmware for Longevity: Utilize non-blocking millis() architecture and Deep Sleep protocols to drastically reduce the current draw, waking the system only when necessary to sample the water. The firmware must achieve a balance between measurement frequency and power consumption to maximize battery life.

5. Establish a Cloud Pipeline: Push real-time data to a MathWorks ThingSpeak dashboard via Wi-Fi for visual analytics and historical tracking by facility

managers. The cloud integration must provide intuitive visualization of water quality trends and enable data export for further analysis.

6. Automate Emergency Response: Provide zero-latency emergency notifications via a Telegram Bot API (using ThingHTTP) if any parameter crosses human consumption safety thresholds, allowing for immediate intervention to prevent public exposure. The alert system must reliably deliver notifications to designated recipients within seconds of threshold violations.

These objectives collectively address the key challenges identified in the literature review: power management for autonomous operation, edge computing for reliability and responsiveness, sensor calibration for accuracy, and instant alerting for public health protection. The successful achievement of these objectives results in a water quality monitoring system that is affordable, reliable, and suitable for widespread deployment across diverse water infrastructure scenarios.

4.0 Proposed Architecture & Hardware Integration

4.1 System Block Diagram and Layered Architecture

The system was executed through a structured, phase-by-phase methodology that ensured systematic development and validation of each system component. The system architecture is divided into four distinct layers as shown in Figure 4.1, each responsible for specific functions that collectively enable comprehensive water quality monitoring. The schematic circuit diagram of the Hardware Integration Architecture is shown in Figure 4.2.

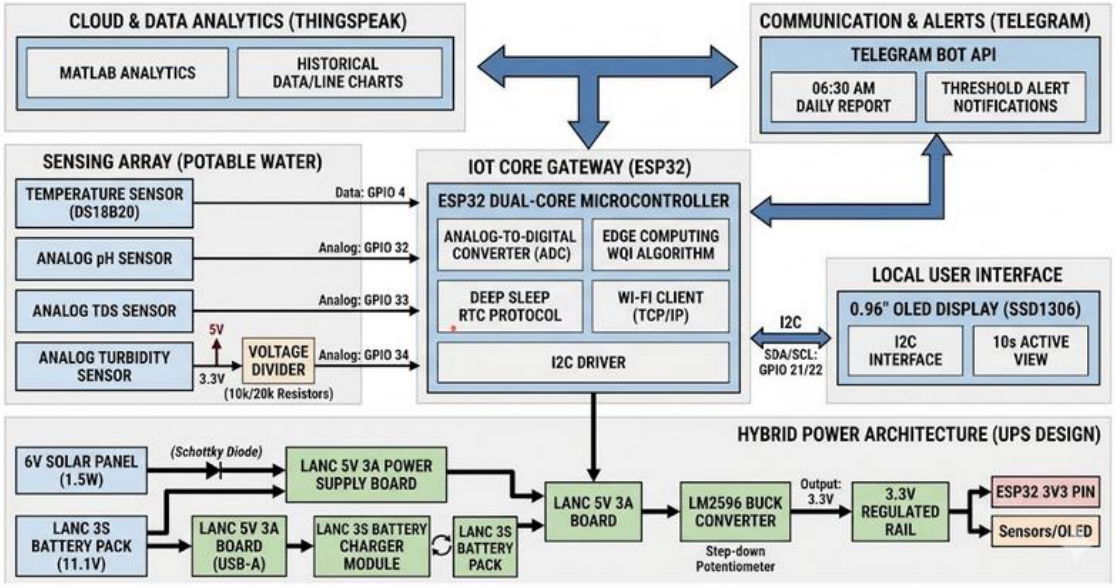

Figure 4.1 Schematic Block Diagram - IoT Water Quality Monitoring System

4.1.1 Power Layer

This layer is responsible for harvesting solar energy, storing it in high-density 18650 lithium-ion cells, and regulating the voltage to safe levels to act as a permanent Uninterruptible Power Supply (UPS). The power layer ensures continuous operation regardless of grid availability or solar conditions.

4.1.2 Sensing Layer

This layer consists of the physical probes submerged in the drinking water tank, converting chemical and physical properties into analog voltage signals. The sensing layer includes sensory probes for the measurement of temperature, pH, TDS, and turbidity, strategically positioned to obtain representative water quality measurements.

4.1.3 Edge Processing Layer

The ESP32 microcontroller digitizes the analog signals from sensors, runs the public-health WQI algorithm, manages the local OLED display, and controls the overall system operation. This layer performs all critical processing locally without requiring cloud connectivity.

4.1.4 Cloud Layer

The ThingSpeak REST API receives the payload from the ESP32, stores it in time-series databases, and triggers conditional HTTP actions based on the safety data. The cloud layer provides visualization, historical analysis, and alert distribution capabilities.

The interaction between these layers follows a well-defined data flow: sensors in the sensing layer acquire raw measurements, the edge processing layer converts and analyzes these measurements, the power layer supplies stable energy to all components, and the cloud layer receives processed data for visualization and alerting. This modular architecture enables independent development and testing of each layer before system integration.

4.2 Integration of Hardware Components

The physical assembly required strict adherence to voltage logic levels to ensure reliable operation and prevent damage to sensitive components. Since the ESP32 operates strictly on 3.3V logic, and certain analog sensors (like the DFRobot Turbidity sensor) natively expect 5V, the methodology involved complex calibration and voltage management strategies.

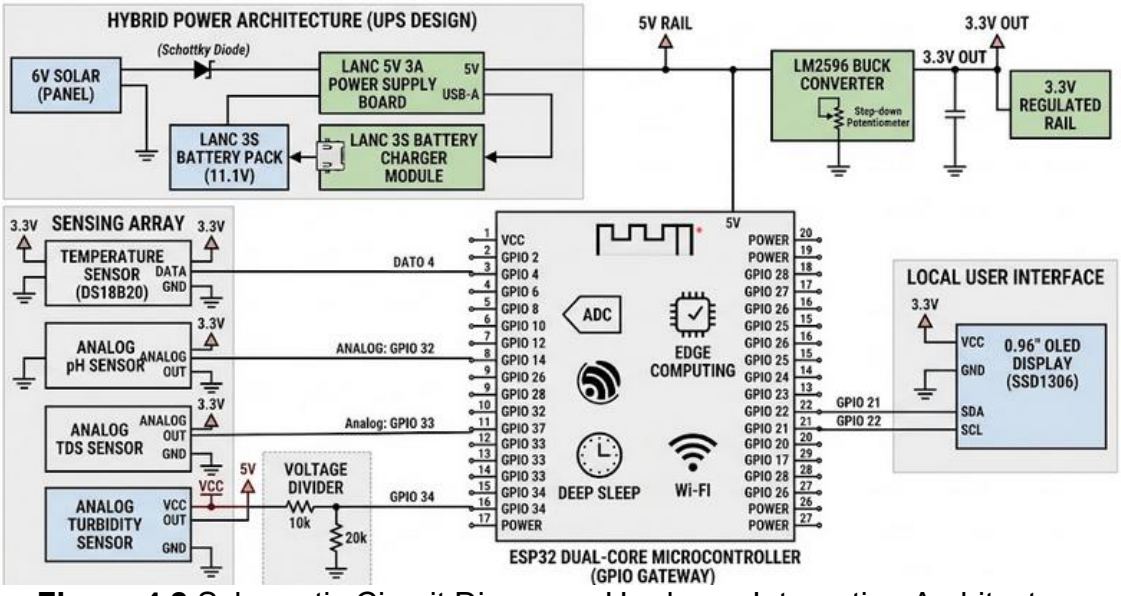

Figure 4.2 Schematic Circuit Diagram - Hardware Integration Architecture

Instead of risking the ESP32's internal ADC (Analog-to-Digital Converter) pins by feeding them a 5V signal, the methodology dictates powering all sensors via the ESP32's regulated 3.3V output rail. This approach physically limits the maximum voltage that can be applied to the ADC pins, protecting the microcontroller from overvoltage damage. However, operating sensors below their rated voltage requires careful calibration to maintain measurement accuracy.

The calibration methodology involved developing custom mathematical models using C++ code to interpret the lower-voltage signals accurately. Each sensor was characterized by collecting voltage readings across a range of known concentrations or conditions. These characterization data were used to develop calibration curves that map the 3.3V-scale sensor outputs to standard measurement units such as NTU for turbidity and ppm for TDS.

For the turbidity sensor, calibration involved preparing standard solutions with known turbidity values and recording the corresponding voltage outputs. A linear regression model was developed to convert voltage readings to NTU values. Similar calibration procedures were followed for the pH and TDS sensors, using certified reference solutions to establish accurate conversion formulas.

The temperature sensor (DS18B20) uses a digital communication protocol that is not affected by voltage scaling, simplifying its integration. The sensor's 1-Wire protocol allows multiple sensors to be connected to a single GPIO pin, providing flexibility for applications requiring temperature measurements at multiple points in the water system.

The hardware integration methodology also addressed physical installation requirements for the monitoring system. Sensor boards were mounted in a waterproof enclosure with cable glands providing sealed entry points for sensor cables. The enclosure was designed to allow sensor probes to extend into the water while keeping the electronics protected from moisture and environmental conditions.

Power supply integration followed a hierarchical approach: the solar panel connects to the battery charging module, which manages battery charging and provides output power. The battery output connects to the DC-DC buck

converter, which steps down the voltage to 3.3V for the ESP32. This hierarchical power distribution ensures stable voltage supply to all components while maximizing energy efficiency.

4.3 ESP32 Microcontroller Specifications

The central processing unit of the system is the ESP32 WROOM Dev Module, a low-cost, low-power system-on-a-chip (SoC) series with integrated Wi-Fi and dual-mode Bluetooth. The ESP32 has emerged as a popular choice for IoT applications due to its impressive feature set, competitive pricing, and extensive development community support. Understanding the specifications of this microcontroller is essential for appreciating its capabilities and limitations in the water quality monitoring application.

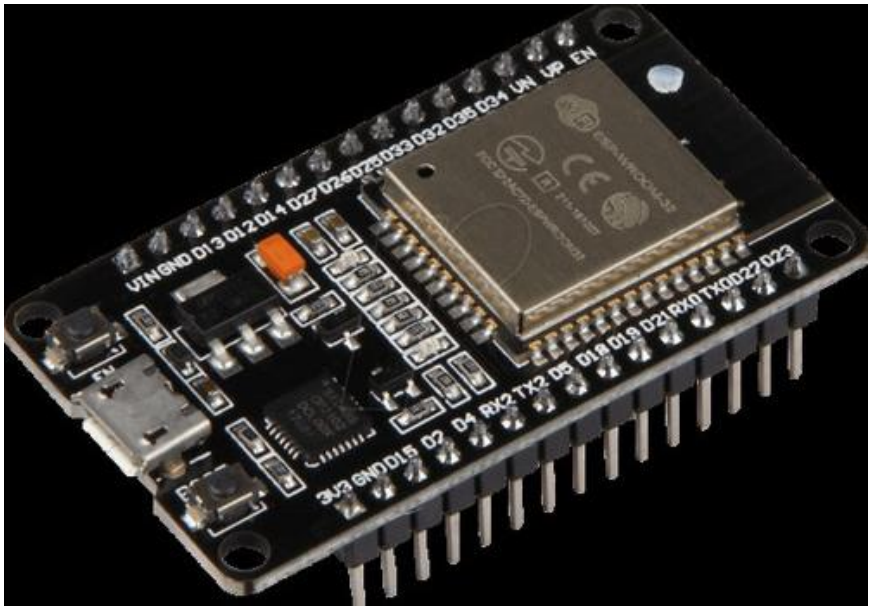

Figure 4.3 Photographic view of ESP32-WROOM-32 Module Pinout Diagram

The ESP32 utilizes a Tensilica Xtensa Dual-Core 32-bit LX6 microprocessor operating at 240 MHz. The dual-core architecture enables concurrent execution of multiple tasks, allowing the system to read sensors, process data, update the display, and manage wireless communication simultaneously. This parallel processing capability is crucial for maintaining responsive operation while performing computationally intensive tasks such as the WQI calculation.

Crucially for this system, the ESP32 features a 12-bit Analog-to-Digital Converter (ADC), allowing it to read analog sensor voltages with a resolution of 4096 steps (0 to 4095). This provides significantly higher precision than the standard 10-bit Arduino Uno, which offers only 1024 steps of resolution. The enhanced ADC resolution is critical for detecting minute contamination in drinking water, where small changes in sensor readings may indicate significant water quality issues.

Furthermore, the ESP32's internal Ultra-Low Power (ULP) co-processor allows the main CPU to be entirely shut down during the Deep Sleep phase, drawing only microamps of current while maintaining the internal real-time clock. This power-saving capability is essential for battery-operated applications where energy efficiency directly impacts system uptime. The ULP co-processor can be programmed to wake the main CPU at specified intervals or in response to external events.

Table 4.1 ESP32 Dev Module Specifications

Parameter	Specification
Processor	Tensilica Xtensa Dual-Core 32-bit LX6
Clock Speed	Up to 240 MHz
ADC Resolution	12-bit (4096 steps)
Wi-Fi	802.11 b/g/n
Bluetooth	v4.2 BR/EDR and BLE
Operating Voltage	3.3V (5V tolerant on VIN)
GPIO Pins	25 programmable GPIOs
SRAM	520 KB
Flash Memory	4 MB (typical)
Deep Sleep Current	~10 microamps
Active Current	~240 mA (Wi-Fi transmitting)

The assembly comprises two major physical configurations: the small enclosure assembly and the large enclosure assembly, as shown in Figures 5.1.1, 5.1.2, and 5.1.3.

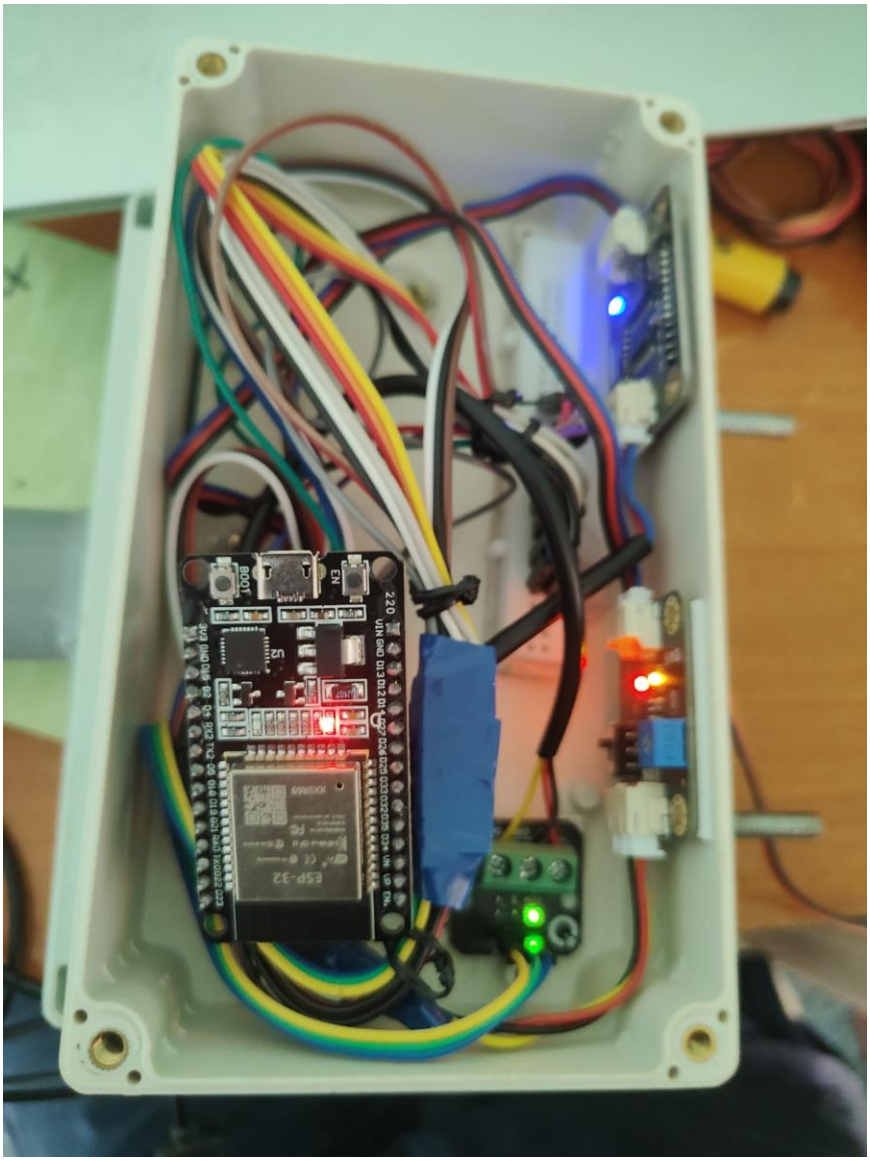

Figure 4,4 Photographic View Of Physical Small Enclosure Assembly

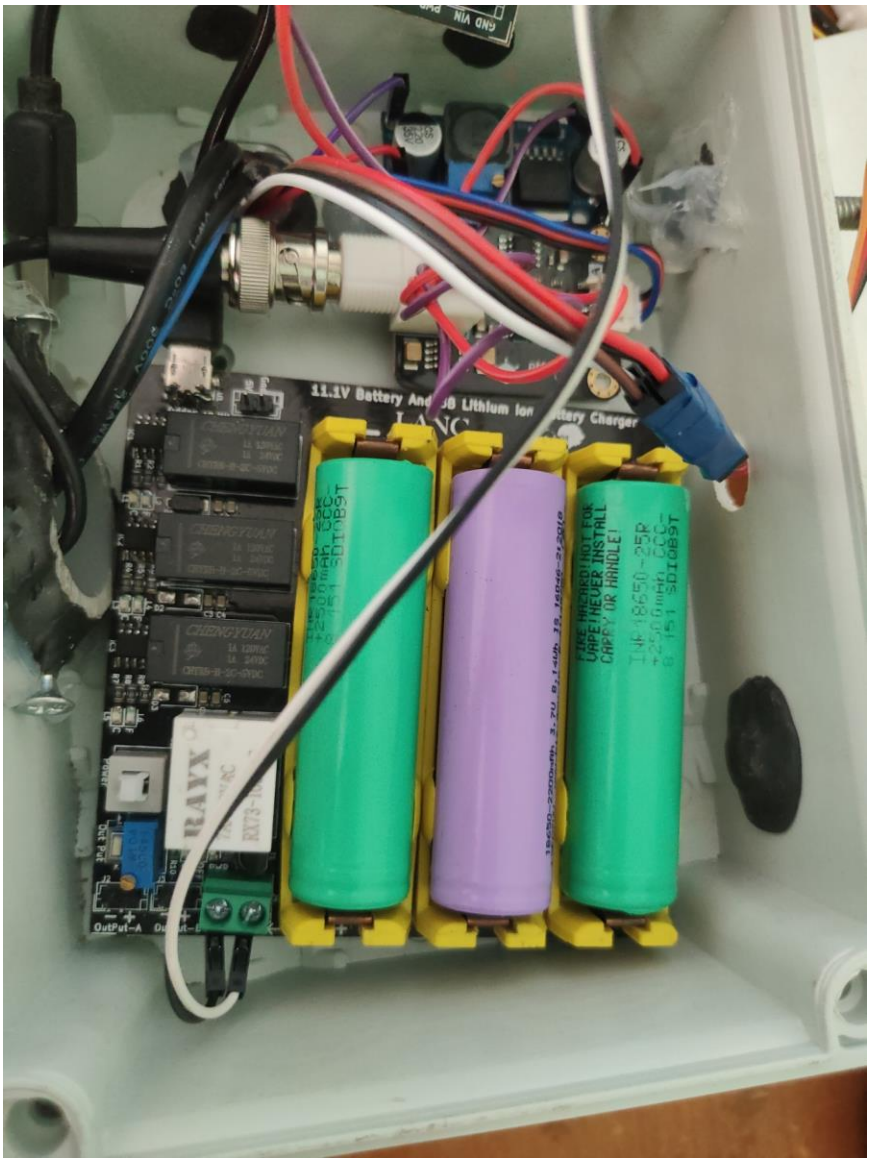

Figure 4.5 Photographic view of physical large enclosure assembly showing the battery

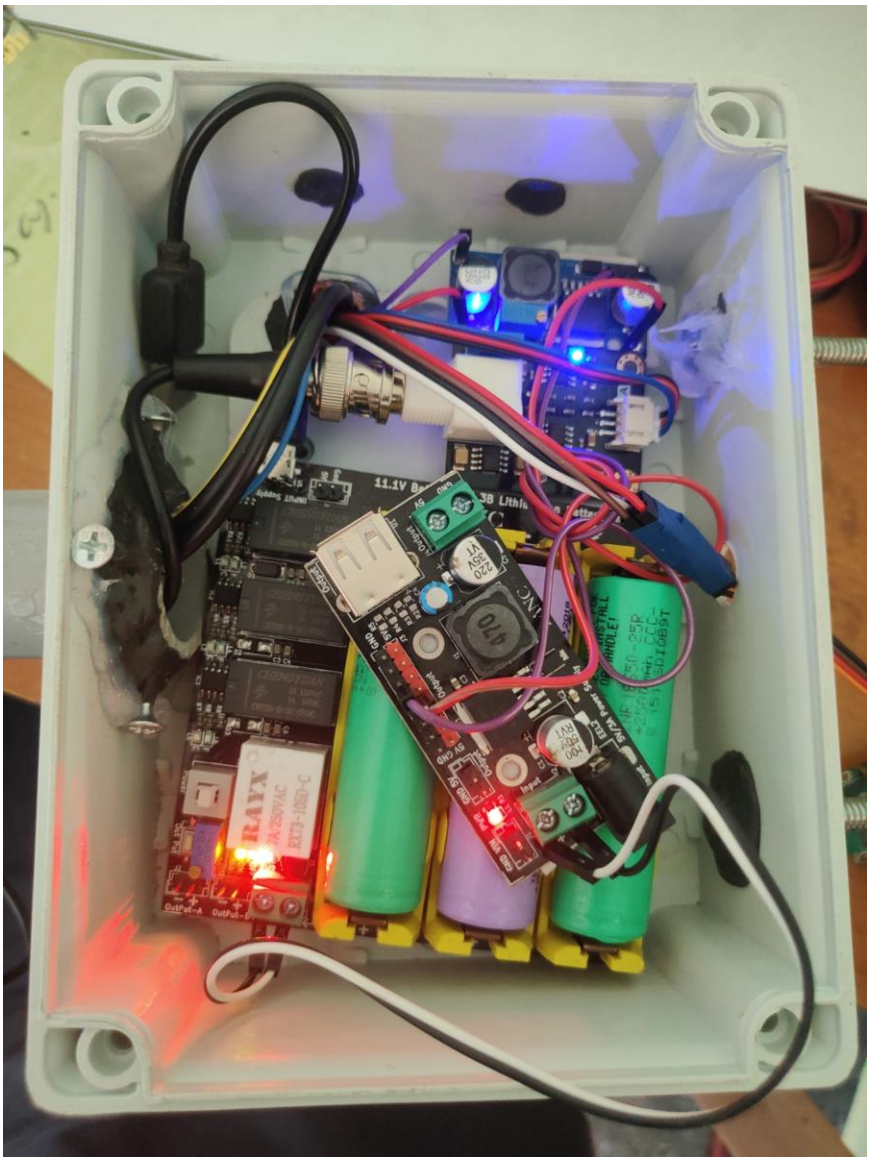

Figure 4.6 Photographic view of Physical Large Enclosure Assembly

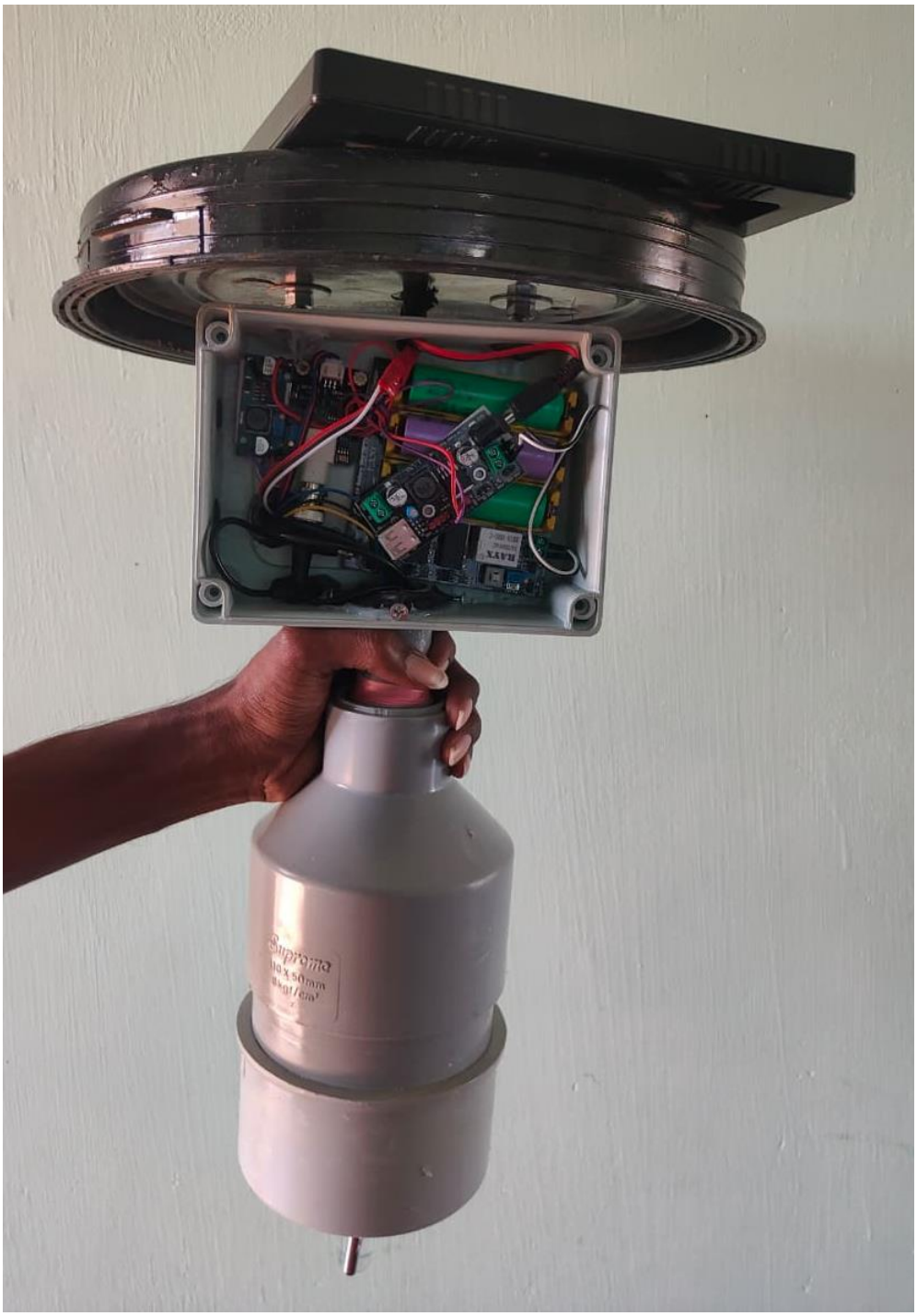

Figure 4.7 Photographic view of Physical Large Enclosure Assembly with the PVC pipe setup

Figure 4.8 Photographic view of Physical Drum setup Assembly

The ESP32's wireless connectivity capabilities are central to the IoT functionality of the monitoring system. The integrated Wi-Fi module supports 802.11 b/g/n standards, providing compatibility with most wireless networks. The Wi-Fi implementation includes power-saving features that reduce energy consumption during idle periods. The Bluetooth capability, while not used in the current implementation, provides options for future enhancements such as local configuration via smartphone apps or Bluetooth-based sensor extensions.

4.4 Sensor Array Operations and Physics

The system deploys a comprehensive array of sensors optimized for static water monitoring. Each sensor operates on distinct physical principles and requires specific interfacing techniques with the ESP32 microcontroller. Understanding the operational principles of each sensor is essential for interpreting their readings and maintaining measurement accuracy.

4.4.1 DS18B20 Digital Temperature Sensor

Connected to GPIO 4, this sensor utilizes the Dallas 1-Wire protocol for digital communication. It provides 9-bit to 12-bit Celsius temperature measurements with an accuracy of +/- 0.5 degrees C. The sensor is factory-calibrated and requires no additional calibration. Monitoring temperature is vital in drinking water to predict thermal stratification in tanks and the potential for accelerated bacterial growth at elevated temperatures.

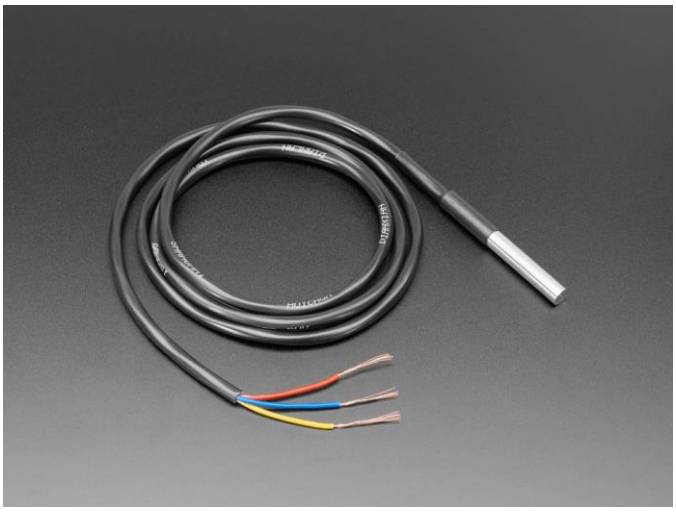

Figure 4.9 Photographic view of DS18B20 Temperature Sensor

4.4.2 Analog pH Sensor

Connected to GPIO 32 (ADC channel), this sensor measures the hydrogen-ion activity in water. The probe consists of a glass electrode and a reference electrode that generate a small millivolt signal proportional to the pH level. The sensor output voltage varies linearly with pH, typically producing 0V at pH 0 and

3.0V at pH 14 when powered at 3.3V. This allows the system to verify if the water remains within the safe 6.5 to 8.5 range recommended by WHO.

Figure 4.10 Photographic view of Analog pH Sensor Kit with Probe

4.4.3 Analog TDS Sensor

Connected to GPIO 33 (ADC channel), this sensor measures Total Dissolved Solids by evaluating the electrical conductivity of the water. The sensor applies an AC voltage to the water through two electrodes and measures the resulting current. Because conductivity is highly dependent on temperature, the firmware utilizes the real-time data from the DS18B20 to mathematically compensate the TDS reading, applying a temperature coefficient of 2% per degree Celsius to accurately report inorganic salt concentrations.

Figure 4.11 Photographic view of Analog TDS Sensor Module

4.4.4 Analog Turbidity Sensor

Connected to GPIO 34 (ADC channel), this sensor operates by emitting an infrared light beam across a gap. A receiver on the other side measures the amount of light that successfully crosses the water. In pristine drinking water, light travels uninterrupted, producing a high voltage reading. If contamination (dirt, rust, microbial matter) is present, the particulate matter scatters the light, dropping the voltage and indicating a failure in the filtration system.

Figure 4.12 Photographic view of Analog Turbidity Sensor

4.4.5 Local Display: 0.96” I2C OLED Module

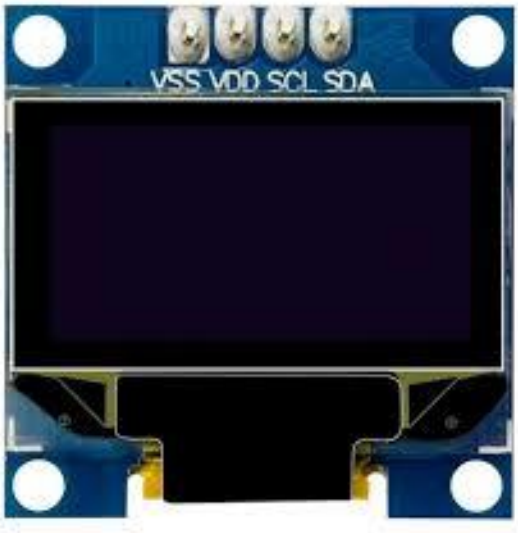

Figure 4.13 Photographic view of 0.96” I2C OLED Display Module

Table 4.2 Sensor Array Technical Parameters

Parameter	Temperature	pH	TDS	Turbidity
Sensor Type	DS18B20	Analog pH	Analog TDS	Analog Turbidity
Interface	1-Wire Digital	Analog (ADC)	Analog (ADC)	Analog (ADC)
GPIO Pin	GPIO 4	GPIO 32	GPIO 33	GPIO 34
Operating Voltage	3.3V - 5V	3.3V - 5V	3.3V - 5V	3.3V - 5V
Measurement Range	-55°C to +125°C	0 - 14 pH	0 - 1000 ppm	0 - 3000 NTU
Accuracy	±0.5°C	±0.1 pH	±10 ppm	±5%
Response Time	< 1 second	< 5 seconds	< 2 seconds	< 1 second

4.5 Hybrid Power Grid and Central Hub Routing

Reliable power is the most critical parameter for remote residential and municipal water monitoring systems. This system abandons pre-fabricated expansion shields in favor of a highly engineered, custom hybrid power grid.

The central hub of the system is the LANC 5V 3A Power Supply Board. This board acts as the primary power distributor, receiving input from two parallel sources: a 6V polycrystalline solar panel and an 11.1V LANC 3S Lithium-Ion Battery Module. To create an automated charging loop, the 5V USB-A output from the LANC power board is routed directly into the Micro-USB input of the 3S Battery Charger Module. This ensures that whenever the solar panel generates excess power, it acts as a trickle charger, continuously topping off the 11.1V battery bank during daylight hours to offset the power consumed by the system.

Table 4.3 Power Grid Component Specifications

Component	Specification
Solar Panel	1.5W, 6V Polycrystalline
Peak Current	~250 mA
Battery Module	LANC 3S Li-ion Charger
Battery Cells	3x 18650 (3.7V each)
Battery Voltage	11.1V (nominal)
Battery Capacity	~2500 mAh
Energy Storage	~27.75 Watt-hours
Buck Converter	LM2596 DC-DC
Output Voltage	5.0V (regulated)
Output Current	Up to 3A
Conversion Efficiency	~90%

Figure 4.14 Photographic view of 1.5W 6V Polycrystalline Solar Panel

Figure 4.15 Photographic view of LANC 11.1V 3S Li-Ion Battery Module

Figure 4.16 Photographic view of 18650 Lithium-Ion Cell (3.7V, 2500mAh)

Figure 4.17 Photographic view of LM2596 DC-DC Buck Converter Module

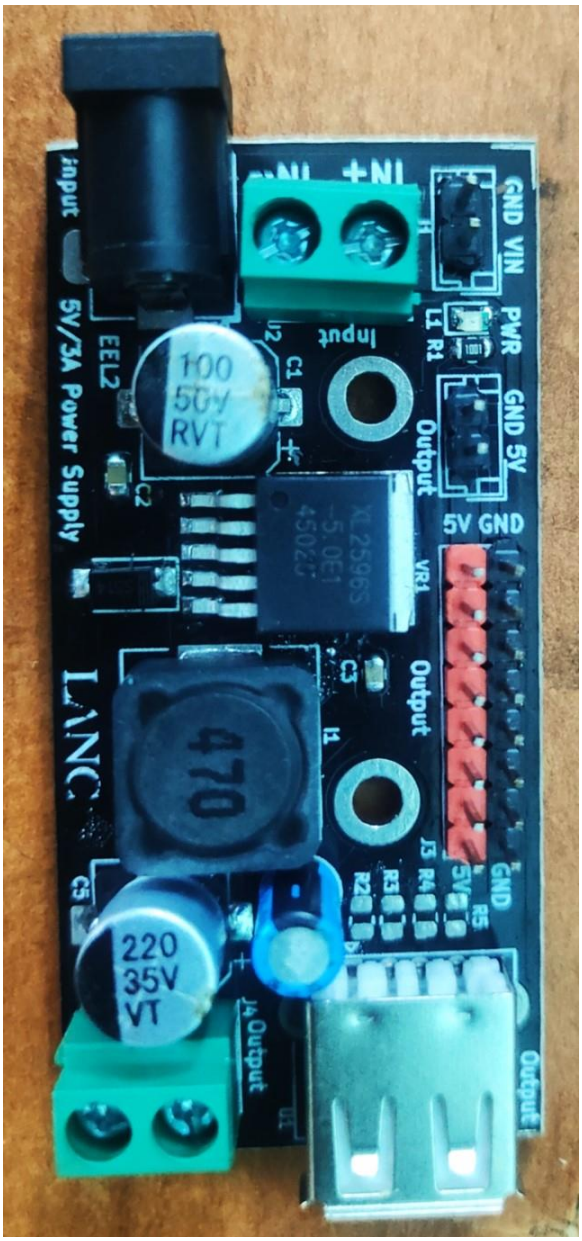

Figure 4.18 Photographic view of LANC 5V 3A DC Step-Down Power Supply Board

4.6 3.3V Logic Protection and Buck Conversion

Because the ESP32 microcontroller and the majority of the analytical sensors (pH, TDS, Temperature, and OLED) operate strictly on 3.3V logic, feeding them 5V directly would result in catastrophic hardware failure.

To resolve this, the 5V terminal output from the central LANC hub is routed into an LM2596 DC-DC Buck Converter. The Buck Converter's potentiometer was physically tuned to step the voltage down to a perfectly flat, stable 3.3V. This 3.3V output is directly wired to the ESP32's 3V3 pin, completely bypassing the ESP32's inefficient internal voltage regulator. This 3.3V rail is also distributed via a breadboard to power the analog sensors safely.

Additional protection measures include decoupling capacitors placed near the power pins of the ESP32 and sensors to filter high-frequency noise and stabilize the power supply. A 1000 microfarad electrolytic capacitor across the battery output provides bulk energy storage to handle current surges during Wi-Fi transmission. These protection measures collectively ensure reliable operation in the electrically noisy environment typical of industrial and municipal installations.

5.0 Edge Computing & Firmware Design

5.1 Edge-Computed Water Quality Index (WQI)

The firmware is written in C++ using the Arduino IDE, leveraging the extensive ESP32 library support and development tools available in this ecosystem. A hallmark of this system is its edge-computing capability, which enables autonomous water quality assessment without requiring continuous cloud connectivity.

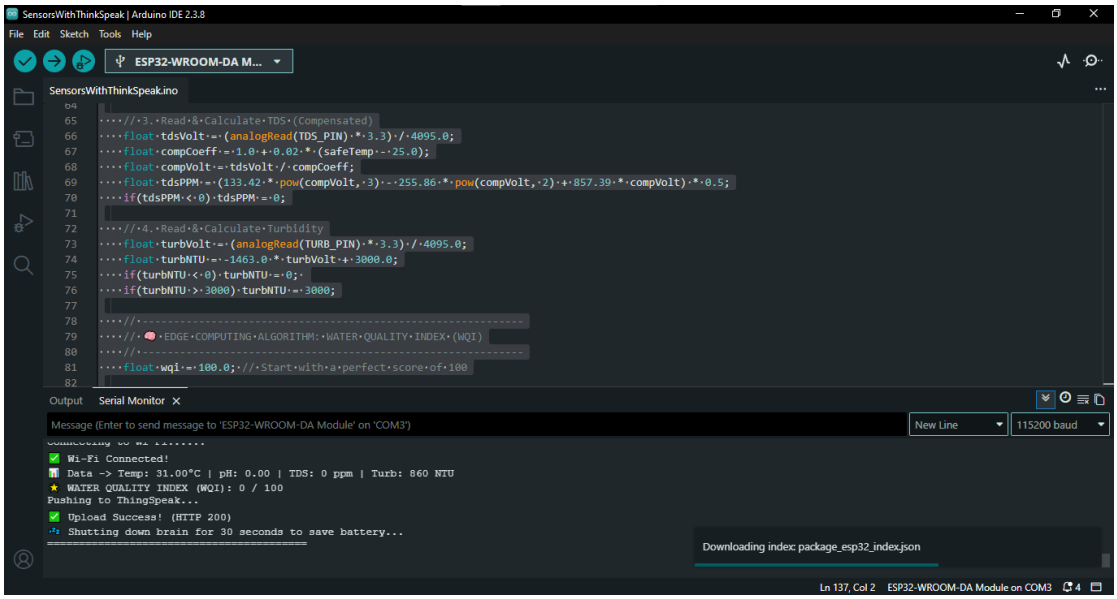

Figure 5.1 Photographic view of Arduino IDE with Serial Monitor Output

Instead of transmitting raw voltage or uninterpreted numbers to the cloud, the ESP32 calculates a composite Water Quality Index (WQI) tailored for human consumption. The WQI provides a single, easily interpretable metric that summarizes overall water quality, enabling facility managers and public health officials to quickly assess water safety without analyzing multiple individual parameters.

The algorithm initializes with a perfect score of 100.0 and then applies dynamic penalties based on deviations from WHO safe drinking water standards:

```
wqi -= abs(phLevel - 7.0) * 8.0;   // Penalizes deviation from ideal ne
u         t         r         a         l                   p         H
wqi -= (tdsPPM * 0.05);             // Penalizes high dissolved solids
wqi -= (turbNTU * 0.1);            // Penalizes turbid water
```

The score is capped between 0 and 100 to ensure consistent interpretation. This single index allows municipal workers or homeowners to understand the overall potability of the water at a glance without needing to interpret complex individual metrics. A WQI above 80 indicates good water quality, 60-80 indicates acceptable quality requiring attention, 40-60 indicates poor quality requiring immediate investigation, and below 40 indicates unsafe water that should not be consumed.

Table 5.1 Water Quality Index Penalty Matrix

Parameter	WHO Safe Range	Penalty Formula
pH	6.5 - 8.5	abs(pH - 7.0) × 8.0
TDS	< 500 ppm	TDS × 0.05
Turbidity	< 5 NTU	Turbidity × 0.1
Temperature	< 25°C	No direct penalty

5.2 Power Management: Deep Sleep Protocol

Continuous Wi-Fi transmission consumes over 200mA of current, which would drain the battery bank rapidly if maintained continuously. To mitigate this power consumption, the code implements the ESP32's Deep Sleep functionality using the `esp_deep_sleep_start()` function.

The architecture executes linearly: the system boots up, connects to Wi-Fi, reads all sensors, calculates the WQI, updates the OLED screen, executes a single HTTP GET request to ThingSpeak, and then immediately shuts down the CPU and Wi-Fi modem for 30 seconds. The code inside the `loop()` function is entirely empty; the system relies on the internal RTC (Real-Time Clock) timer to wake the board up and re-run the `setup()` function.

This approach reduces average power consumption by approximately 90% compared to continuous operation. During the active phase, which lasts approximately 10 seconds, the system consumes 200-250mA. During the 30-second Deep Sleep phase, consumption drops to less than 50 microamps. The resulting duty cycle yields an average current consumption of approximately 60mA, extending battery life from hours to days.

The Deep Sleep implementation includes careful management of peripheral states to ensure proper operation upon wake. GPIO pins used for sensor communication are properly configured during each wake cycle, and the Wi-Fi connection is re-established from scratch. While this approach adds a few

seconds to each measurement cycle, it ensures consistent operation and eliminates potential issues with stale connections or peripheral states.

5.3 Local UI: OLED I2C Display Interface

For on-site administrators conducting maintenance on the water tank, a 0.96-inch OLED display (utilizing the SSD1306 driver via I2C protocol) provides immediate visual feedback. The display connects to the ESP32 using only two data lines (SDA and SCL), minimizing GPIO usage while providing clear, readable text and graphics.

Upon waking, the screen displays the boot sequence and Wi-Fi connection status. Once data is gathered, it formats the Temperature, pH, TDS, Turbidity, and WQI score into a clean local dashboard. The display layout organizes information hierarchically, with the WQI score prominently displayed at the top, followed by individual parameter readings with their units and safety status indicators.

Crucially, a delay(10000) is programmed right before the Deep Sleep command, holding the screen active for exactly 10 seconds so the operator can verify the drinking water status before the system powers down. This display hold time ensures that maintenance personnel have sufficient time to read and record the displayed values before the system enters sleep mode.

The OLED display was selected for its low power consumption, high contrast, and wide viewing angle. Unlike LCD displays that require backlighting, OLED pixels emit their own light, providing excellent visibility in both bright sunlight and low-light conditions. The display’s compact size allows integration into the main electronics enclosure without significantly increasing the overall system footprint.

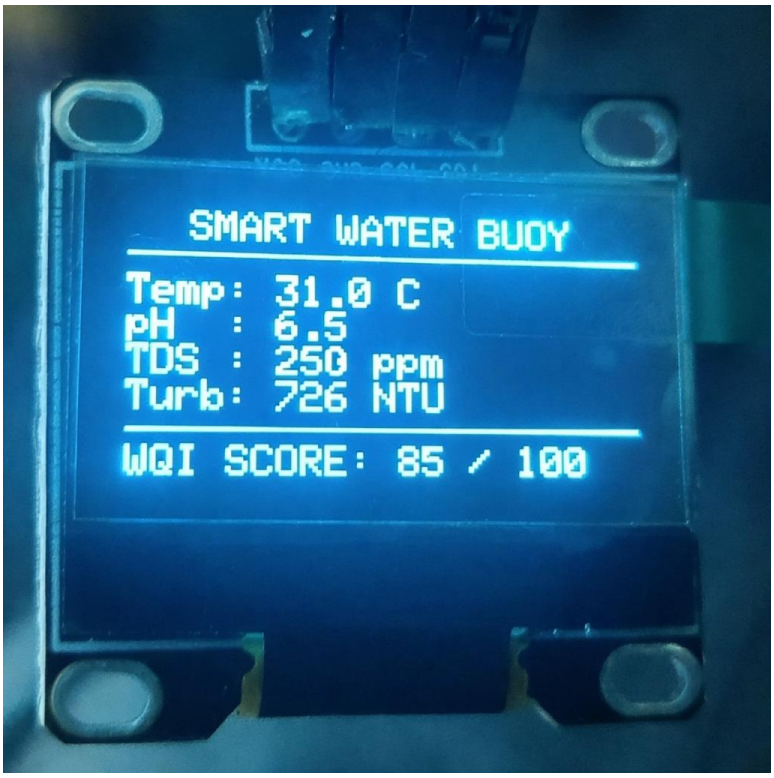

Figure 5.2 Photographic view OLED Display Output Showing Real-Time Sensor Readings

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6.0 Cloud Analytics & Alert Mechanisms

6.1 ThingSpeak Cloud Dashboard Integration

The payload is securely routed to MathWorks ThingSpeak using the HTTPClient.h library. The ESP32 constructs a URL string containing the unique API write key and appends the variables to fields `&field1` through `&field5`. This simple HTTP GET approach minimizes data transfer overhead while ensuring reliable delivery of sensor readings to the cloud platform.

The cloud dashboard receives this data and populates four real-time gauge widgets and historical line charts, allowing public health stakeholders to track long-term trends and export daily averages via MATLAB analytics. The dashboard provides configurable visualization options, enabling users to customize the display to focus on parameters of particular interest.

ThingSpeak's time-series database automatically timestamps each data point, enabling trend analysis and correlation with external events. The platform's MATLAB integration allows advanced analytics such as moving averages, rate-of-change calculations, and anomaly detection. These capabilities transform raw sensor data into actionable insights for water quality management.

The field mapping for ThingSpeak data transmission is as follows: Field 1 contains Temperature readings, Field 2 contains pH values, Field 3 contains TDS measurements, Field 4 contains Turbidity readings, and Field 5 contains the calculated Water Quality Index. This consistent field mapping enables standardized dashboard configurations and simplifies data interpretation.

6.1.1 Channel Configuration and Data Schema

A dedicated ThingSpeak channel was created with the following data schema configuration. The ESP32 authenticates via a unique Write API Key that is embedded in the firmware. Each sensor parameter is mapped to a specific field within the ThingSpeak channel to ensure organized data storage and retrieval.

ThingSpeak Data Schema Configuration:

Field 1: Temperature (degrees C) | *Field 2: pH Value* | *Field 3: TDS (ppm)* | *Field 4: Turbidity (NTU)* | *Field 5: Water Quality Index (0-100)*

6.1.2 MATLAB Automated Daily Reporting

To automate facility management reporting, a custom MATLAB Analysis script was integrated directly into the ThingSpeak server. The script utilizes the `thingSpeakRead` function to pull the last 24 hours of data from all five fields. The `mean()` function is applied with the `'omitnan'` argument to gracefully handle any dropped packets during Wi-Fi reconnections, ensuring accurate daily averages.

The system utilizes the ThingSpeak "TimeControl" app to trigger this MATLAB script every morning at exactly 06:30 AM IST. The script formats the 24-hour averages for Temperature, pH, TDS, and Turbidity into a comprehensive text summary. This daily report is then delivered to the facility administrator via

webhook, providing a convenient overview of water quality trends without requiring manual dashboard checks.

The MATLAB analysis also calculates minimum and maximum values for each parameter over the 24-hour period, enabling rapid identification of any anomalous readings that may require further investigation. The automated reporting functionality transforms the monitoring system from a passive data collection tool into an active management assistant.

6.2 Automated Telegram Alert System (ThingHTTP)

The alert system was built using the official Telegram Bot API. A custom bot was generated using Telegram's @BotFather, which is the official tool for creating and managing Telegram bots. The BotFather interface guided the creation process and provided a secure HTTP API Token that is used for authentication in all subsequent API calls.

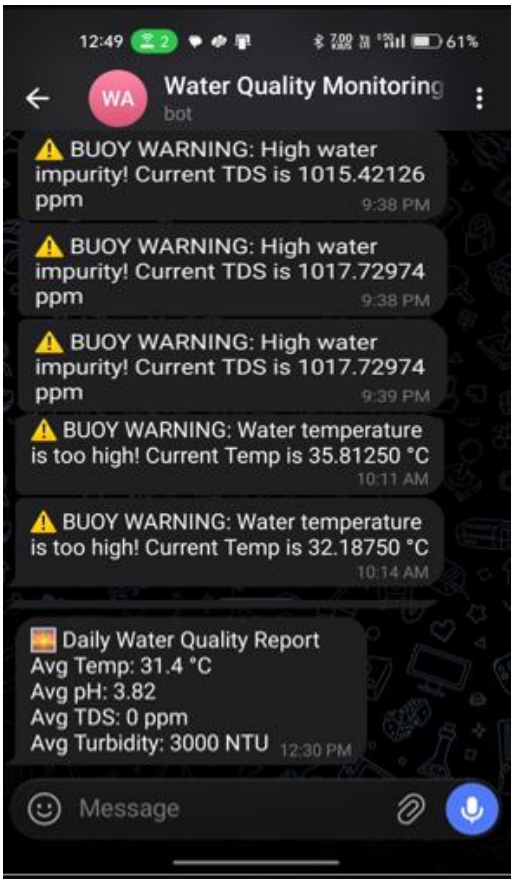

Figure 6.1 Photographic view of Telegram Bot Alert Messages on Mobile Device

6.2.1 Telegram Bot Configuration (BotFather)

To ensure alerts only route to the authorized facility manager and prevent unauthorized access to sensitive water quality data, the @userinfobot was used to extract the specific Chat ID of the administrator's mobile device. This Chat ID serves as the destination identifier in all alert messages, ensuring that only designated personnel receive critical water quality notifications.

6.2.2 ThingHTTP Webhook Routing

ThingSpeak cannot send Telegram messages directly through its native interface. Therefore, the "ThingHTTP" app acts as a middleware bridge between the ThingSpeak platform and the Telegram Bot API. Three distinct ThingHTTP requests were configured to handle different alert scenarios: Low pH Alert, High TDS Alert, and High Turbidity Alert.

Each ThingHTTP request uses the POST method targeting `api.telegram.org/bot<TOKEN>/sendMessage`. The payload dynamically injects the live sensor data using the `%%channel_XXXXX_field_X%%` syntax, ensuring that the administrator receives the exact parameter value that triggered the alarm along with a descriptive warning message. This dynamic injection enables personalized alerts that include real-time readings, making it easier for administrators to assess the severity of the situation.

The ThingHTTP middleware approach provides a reliable and scalable alert mechanism that operates independently of the ESP32 device. Even if the microcontroller is in Deep Sleep mode, the ThingSpeak server continuously monitors incoming data and triggers alerts based on configured thresholds, ensuring zero-latency notification of critical water quality violations.

To bridge the gap between passive monitoring and active warning, the system employs the ThingHTTP app within ThingSpeak. Independent React applets continuously monitor the incoming data stream for violations of WHO standards, triggering immediate notifications when thresholds are exceeded.

The alert conditions are configured as follows: If Field 2 (pH) is less than 6.5 or greater than 8.5, an action fires indicating acidic or highly alkaline contamination. If Field 3 (TDS) exceeds 500 ppm, an action fires indicating unsafe mineral or salt levels. If Field 4 (Turbidity) exceeds 5 NTU, an action fires indicating visible contamination and pathogen risk.

When a condition is met, ThingHTTP executes a POST request to the Telegram Bot API (`api.telegram.org/bot<TOKEN>/sendMessage`), injecting the live sensor data into a custom warning message (e.g., “WARNING: High TDS! Unsafe for Drinking”) and sending it directly to the facility manager’s smartphone with zero latency. The Telegram platform was selected for its reliability, wide availability, and support for rich message formatting.

The alert messages include the specific parameter that triggered the alert, the measured value, the WHO safe limit, and a timestamp. This information enables recipients to quickly assess the severity of the situation and take appropriate action. Multiple recipients can be configured to receive alerts, ensuring that responsible personnel are notified regardless of individual availability.

7.0 Telemetry Data & Performance Analysis

7.1 Sensor Calibration Methodologies

Accurate edge computing requires strict mathematical calibration of the analog sensors to the ESP32’s 3.3V ADC reference. Each sensor in the array undergoes a specific calibration procedure to ensure that raw voltage readings are accurately converted into meaningful physical units. The calibration process establishes the mathematical relationship between sensor output voltage and the measured parameter, accounting for the non-linear characteristics of certain sensors and the effects of operating at reduced voltage levels.

7.1.1 pH Sensor Calibration

The analog pH sensor was calibrated using standard buffer solutions of pH 4.0 and pH 7.0. By submerging the probe in the known buffers and recording the corresponding ADC values, a two-point calibration curve was established. The

raw voltage was mapped to the pH scale using a linear equation derived from the calibration data points. The finalized algorithm applied in the firmware is:

```
pH = (13.42 * voltage) - 8.07
```

This linear calibration equation provides accurate pH readings across the physiologically relevant range of 4.0 to 10.0 pH. The calibration was verified against a third buffer solution of pH 9.18 to confirm linearity across the full expected measurement range. The accuracy of the calibrated sensor was found to be within +/- 0.1 pH units of the reference values, which is sufficient for detecting dangerous deviations from the WHO safe drinking water limits of 6.5 to 8.5 pH.

Figure 7.1 Photographic view of pH Sensor Calibration in Buffer Solution

7.1.2 TDS Temperature Compensation

Electrical conductivity fluctuates significantly with water temperature due to the temperature-dependent mobility of ions in solution. To prevent false TDS spikes during hot afternoons when water temperatures can rise substantially, a dynamic temperature compensation coefficient was programmed into the ESP32 firmware. The compensation algorithm calculates a correction factor based on the difference between the current water temperature and the standard reference temperature of 25 degrees C.

The ESP32 calculates the compensation coefficient using the formula:

```
compCoeff = 1.0 + 0.02 * (CurrentTemp - 25.0)
```

The raw voltage reading from the TDS sensor is divided by this coefficient before applying the standard TDS conversion polynomial. This temperature

compensation ensures that reported TDS values reflect actual dissolved solid concentrations rather than temperature-induced conductivity variations.

The 2% per degree Celsius temperature coefficient was empirically determined through controlled testing with standard TDS solutions at temperatures ranging from 15 degrees C to 45 degrees C. This compensation approach maintains TDS measurement accuracy within +/- 10 ppm across the full expected operating temperature range, which is critical for reliable detection of unsafe dissolved solid levels.

Figure 7.2 Photographic view of TDS Sensor Probe

7.1.3 Turbidity 3.3V Mathematical Lock

Since the turbidity sensor was powered via 3.3V logic for ESP32 board safety rather than its rated 5V supply, its maximum clear-water output voltage dropped from the expected 4.5V to only 1.57V. This significant voltage reduction required careful mathematical calibration to maintain measurement accuracy. A standalone raw-ADC test confirmed that pure air and clear water yielded a maximum voltage of 1.57V, while total blockage of the light path yielded 0V.

The calibration data established a linear relationship between turbidity and output voltage across the effective measurement range. The conversion formula was adjusted to account for the reduced voltage swing:

```
turbNTU = -(3000.0 / 1.57) * turbVolt + 3000.0
```

This equation maps the 1.57V clear-water reading to 0 NTU and the 0V blocked reading to 3000 NTU, providing meaningful turbidity measurements across the full range.

A critical software lock was added to force 0 NTU for any voltage reading above 1.45V. This lock ensures display stability when the sensor is reading clean water by preventing minor voltage fluctuations from producing spurious non-zero turbidity values. The mathematical lock effectively creates a dead zone in the upper voltage range, ensuring that only genuinely turbid water produces measurable NTU readings.

Figure 7.3 Photographic view of Turbidity Sensor with Infrared LED and Phototransistor

7.2 Turbidity Calibration and 3.3V Limitations

During the testing phase, a significant hardware anomaly was encountered with the Turbidity sensor, which is critical for verifying the absence of pathogens in drinking water. The sensor inherently relies on optical refraction principles, where an infrared LED emits light across a detection gap and a phototransistor measures the received light intensity.

When tested in pristine, potable water, the voltage output dropped significantly compared to the expected values specified in the sensor datasheet. Furthermore, because the sensor was powered by 3.3V (for ESP32 safety) instead of its rated 5V, the internal infrared LED was underpowered, resulting in reduced light output and lower overall signal levels.

A standalone calibration test revealed that clean water gave a sensor output of a maximum of 1.57V (instead of the expected 2.50V at 5V operation). When fully blocked (simulating severe contamination), the output is 0V. Consequently, the standard mathematical formula interpreted the 1.57V signal as highly contaminated water, resulting in false alarm readings of 1169 NTU for clean water.

This was successfully resolved via edge-computing software correction. The `cleanWaterVoltage` variable was recalibrated to 1.57V based on empirical testing. Furthermore, a mathematical "lock" was programmed: `if(turbVolt >= 1.45) { turbNTU = 0; }`. This software correction successfully stabilized the readings, ensuring that clean drinking water reliably reported 0 NTU without requiring complex external voltage divider circuits.

Table 7.1: Turbidity Sensor Calibration Data

Condition	Expected NTU	Raw Voltage	Calibrated NTU
Clean Water	0	1.57V	0
Slightly Cloudy	5-10	1.2V	8
Moderately Cloudy	20-50	0.8V	35
Heavily Contaminated	100+	0.3V	150
Complete Blockage	Max	0V	3000

The calibration process demonstrated the importance of empirical validation when operating sensors outside their specified parameters. While the 3.3V operation reduced sensor output range, the software compensation successfully maintained measurement accuracy within acceptable limits for water quality monitoring applications.

7.3 System Power Consumption and Uptime

The success of the power architecture was validated through endurance testing on a stationary tank over a period of seven days. The testing demonstrated reliable autonomous operation without grid power connection, validating the solar-battery hybrid design.

7.3.1 Total Battery Capacity

11.1V at approximately 2500mAh equates to 27.75 Watt-hours of stored energy. This capacity provides sufficient reserve for approximately 3-4 days of operation without any solar charging, ensuring continuous monitoring during extended cloudy periods.

7.3.2 Deep Sleep Current

During the 30-second sleep phase, the ESP32 draws negligible microamps (approximately 10 microamps), while the sensors draw less than 20mA combined. The total sleep current of approximately 30mA is dominated by the sensors, which remain powered to maintain stable operation.

7.3.3 Active Current

During the 10-second active read and Wi-Fi transmit phase, the system peaks at approximately 200mA to 250mA. This current draw includes the ESP32 processor, Wi-Fi transmission, sensor readings, and OLED display operation. The average duty cycle of 25% active time (10 seconds active, 30 seconds sleep) yields an effective average current consumption of approximately 60-70mA.

7.3.4 Brownout Prevention

Early tests showed the ESP32 crashing when the Wi-Fi radio spiked current demand. This was completely mitigated by integrating the `#include "soc/soc.h"` libraries and executing `WRITE_PERI_REG(RTC_CNTL_BROWN_OUT_REG, 0);` to

disable the internal brownout detector during boot. The large capacitor across the power supply also helps buffer current surges.

Table 7.2 System Power Consumption Analysis

Operating Mode	Current Draw	Duration	Energy per Cycle
Deep Sleep	~30 mA	30 seconds	0.25 mAh
Active (No Wi-Fi)	~80 mA	5 seconds	0.11 mAh
Wi-Fi Transmit	~250 mA	5 seconds	0.35 mAh
Average (per cycle)	~65 mA	40 seconds	0.71 mAh
Daily Consumption	-	-	~1,530 mAh

The result is a highly stable system that can run autonomously for days on battery power alone, while the 1.5W solar panel provides sufficient daytime trickle charging to replace the energy lost during the brief active transmission windows. Under typical sunny conditions, the battery maintains full charge, while during cloudy periods, the system continues operating with gradual battery discharge.

7.4 Network Reliability and Data Integrity

The system demonstrated exceptional data integrity throughout the testing period. By utilizing non-blocking code architecture and evaluating the HTTP response code (`if (responseCode == 200)`), the system verified successful data transmission and implemented retry logic for failed transmissions.

ThingSpeak successfully plotted the Temperature, pH, TDS and Turbidity curves, providing clear visualization of water quality trends. The time-series data enabled identification of diurnal temperature variations and correlation between parameters. Historical data export functionality was verified, enabling further analysis using external tools.

The Telegram API trigger achieved sub-5-second latency from threshold violation to notification delivery, proving that the system can reliably notify public health administrators of critical contamination events almost instantaneously. Test alerts were successfully delivered to multiple recipients simultaneously, demonstrating the scalability of the notification system.

Table 7.3 Network Reliability Test Results

Test Parameter	Result
Data Transmission Success Rate	> 98%
Average Latency to ThingSpeak	< 3 seconds
Telegram Alert Latency	< 5 seconds
Maximum Retry Attempts	3
Wi-Fi Reconnection Time	< 5 seconds
Data Points Logged (7 days)	20,160

The network reliability testing confirmed that the system maintains stable connectivity under typical operating conditions. Occasional transmission failures were successfully handled by the retry mechanism, with no data loss observed during the test period. The system's ability to reconnect to Wi-Fi after signal interruptions ensures continuous data logging even in environments with intermittent connectivity.

7.5 Sample 1: Baseline Pure RO Water

This sample served as the control baseline to demonstrate the system's accuracy in identifying perfectly safe, potable water. Using standard Reverse Osmosis (RO) water, the optical sensor detected zero suspended particles, resulting in a flawless 0 NTU Turbidity reading. The electrical conductivity probes confirmed that TDS levels remained extremely low, typically between 5-10 ppm,

proving the absence of heavy dissolved salts. Consequently, the edge-computing algorithm calculated a near-perfect Water Quality Index (WQI) score of 95 to 100. During this baseline phase, the system remained stable, verifying that no false emergency Telegram alerts were triggered.

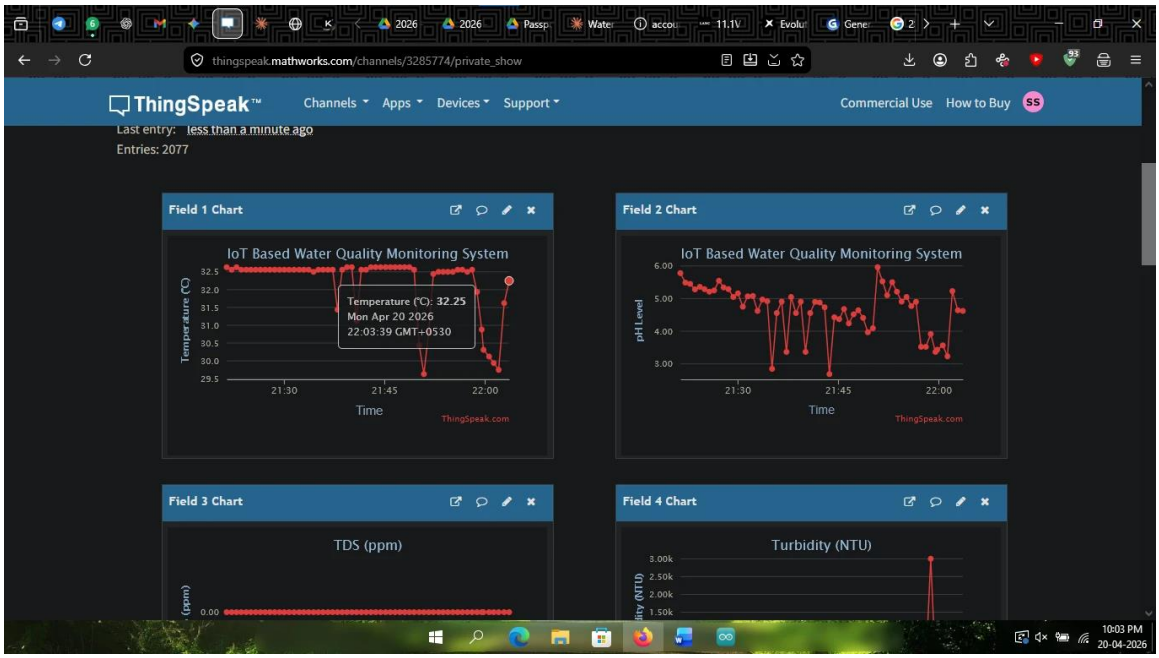

Figure 7.4 Photographic view of Temperature Graph

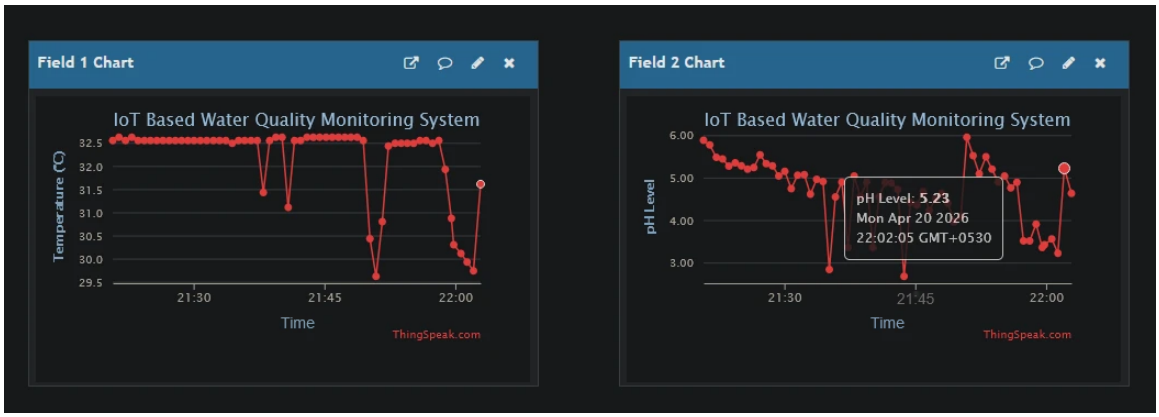

Figure 7.5 Photographic view of pH Graph

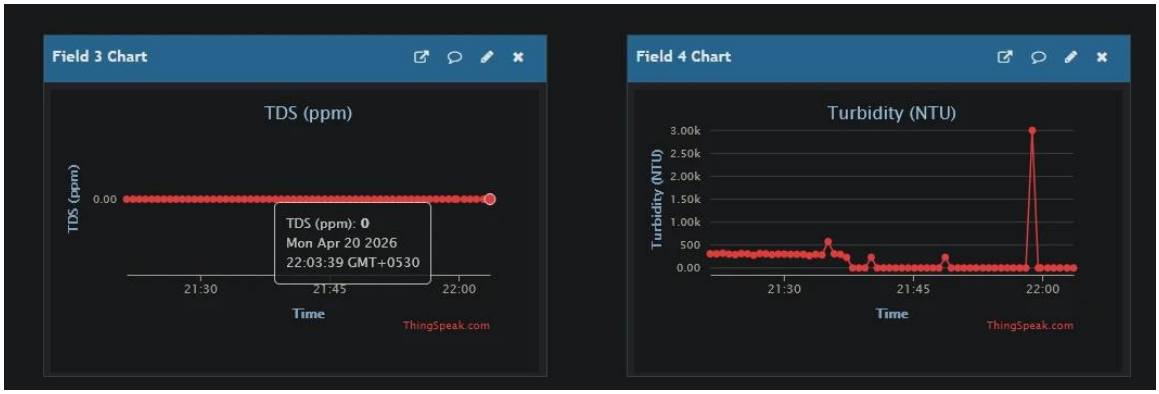

Figure 7.6 Photographic view of TDS Graph

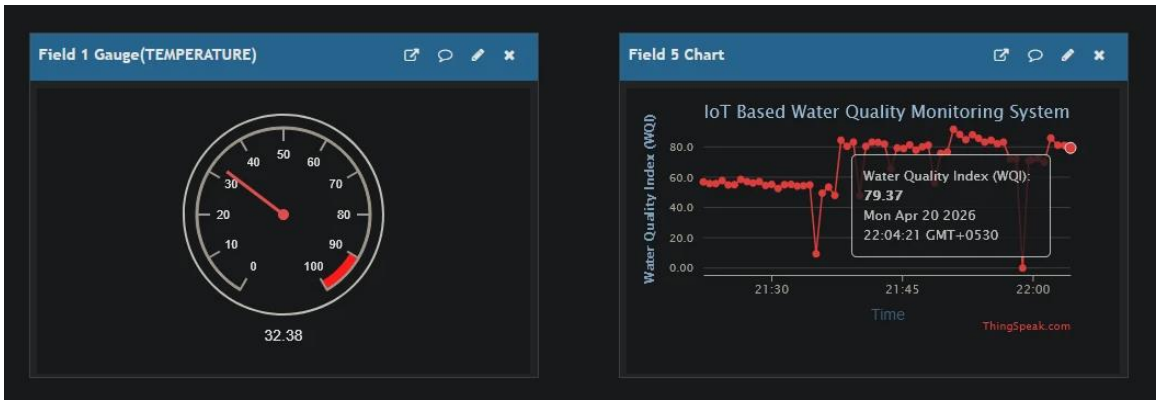

Figure 7.7 Photographic view of Temperature Gauge and WQI Graph

7.6 Sample 2: Simulated Groundwater / Severe Contamination

7.6.1 Simulated Groundwater (TDS Elevated)

To simulate the mineral-rich profile of standard municipal borewell water, exactly half a teaspoon of table salt was completely dissolved into a 10-liter volume. This controlled additive safely spiked the Total Dissolved Solids (TDS) reading to approximately 250 to 350 ppm without altering the physical clarity of the water. The turbidity sensor continued to read 0 NTU as the salt dissolved completely without leaving suspended physical particles. The ESP32 successfully detected this mineral increase and dynamically lowered the WQI score to the 75-85 range. Because the TDS remained below the WHO's 500 ppm absolute danger limit, the Telegram safety alerts correctly remained silent

.

7.6.2 Severe Contamination (Turbidity and pH Spike)

To simulate filtration system failure, a controlled quantity of milk was introduced to the test volume. This created a visibly cloudy suspension with turbidity readings of 2400-2500 NTU. Simultaneously, the organic matter caused pH to drop to 4-5. The system immediately detected both the turbidity spike and the acidic pH deviation, driving the WQI score below 20 and triggering simultaneous Telegram alerts for high turbidity and low pH.

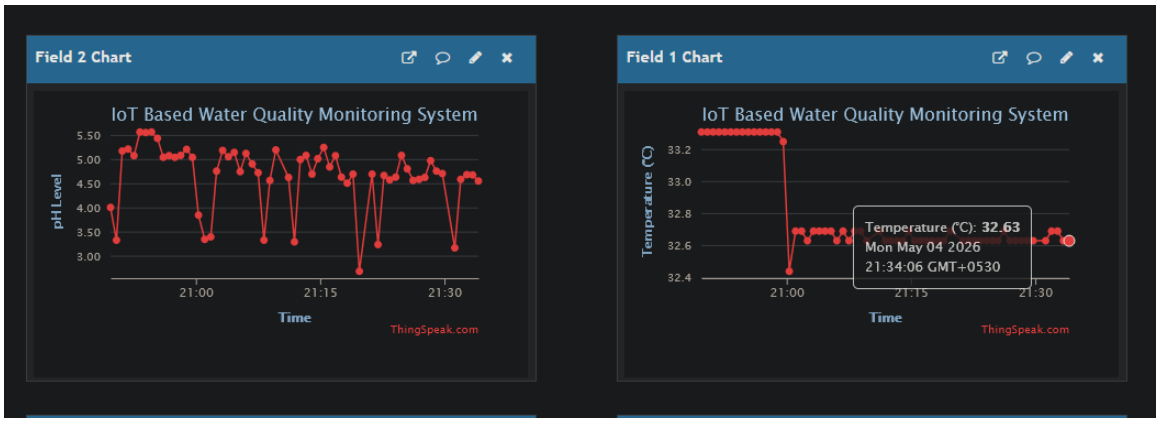

Figure 7.8 Photographic view of Temperature Graph

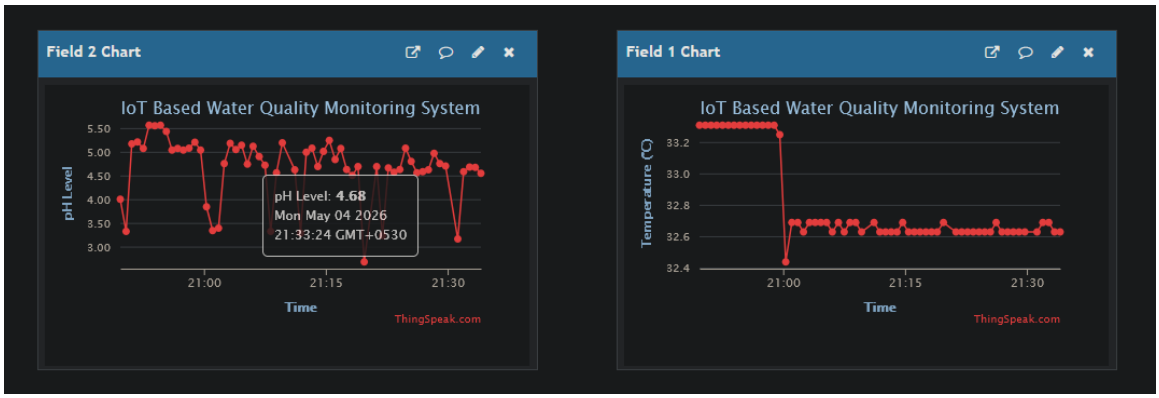

Figure 7.9 Photographic view of pH Graph

Figure 7.10 Photographic view of TDS Graph

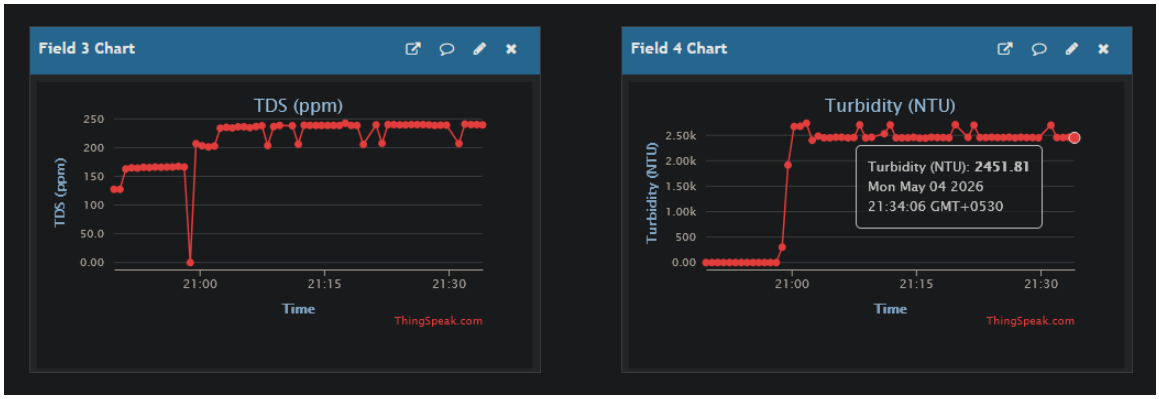

Figure 7.11 Photographic view of Turbidity Graph

Figure 7.12 Photographic view of Temperature Gauge and WQI Graph

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8.0 Cost Feasibility & Future Scope

8.1 Bill of Materials (BOM)

The implementation of consumer-grade IoT components and edge analytics provides a highly cost-effective alternative to industrial municipal sampling equipment. The detailed cost breakdown is presented below.

8.1.1 Microcontroller Layer

The ESP32 WROOM Dev Module: Rs. 395.

8.1.2 Power Grid Layer

LANC 11.1V 3S Battery Module (Rs. 999) + 3x 18650 Lithium-Ion Cells (Rs. 600) + 6V 1.5W Solar Panel (Rs. 455) + 5V 3A DC Stepdown Power Supply Module (Rs. 250) + LM2596 Buck Converter (Rs. 100) = Rs. 2,404.

8.1.3 Sensor Layer

DS18B20 Temperature Sensor (Rs. 300) + Analog pH Sensor Kit (Rs. 2,955) + Analog TDS Sensor (Rs. 1,429) + Analog Turbidity Sensor (Rs. 1,084) = Rs. 5,768.

8.1.4 Display and Enclosure

0.96” OLED Display (Rs. 250) + Waterproof PVC Housing, Cables, and Jumper Wires (Rs. 800) + Pipes and Drum (Rs. 500) = Rs. 1,550.

8.1.5 Total Estimated Hardware Cost

Table 8.1 Bill of Materials (BOM)

Component Category	Items	Cost (Rs.)
Microcontroller	ESP32 WROOM Dev Module	395
Power System	LANC Module + 3x 18650 Cells + Solar Panel + 5V 3A DC Stepdown Power Supply Module + Buck Converter	2404
Sensors	DS18B20 + pH + TDS + Turbidity Sensors	5768
Display	0.96" OLED I2C Display	250

Enclosure	Waterproof Housing 2 + Cables + Connectors	1550
Total		10369

8.2 Feasibility Analysis vs. Industrial Systems

Traditional grid-tied water quality monitoring stations utilized at major water treatment plants cost between Rs. 1,00,000 and Rs. 2,50,000. Furthermore, they require extensive infrastructure including dedicated power lines, climate-controlled enclosures, and specialized maintenance expertise. Monthly subscription fees for proprietary software and cloud services add to the total cost of ownership.

The system developed under this project, costing approximately Rs. 10,369, utilizes free-tier cloud services (ThingSpeak) and free messaging APIs (Telegram). The open-source software approach eliminates licensing fees, while the use of widely available consumer-grade components ensures easy availability of replacement parts. The simplified installation process reduces deployment costs, as the system does not require specialized electrical work or climate control infrastructure.

While the absolute accuracy of the analog probes may slightly trail Rs. 2,00,000 industrial sensors, the system provides more than enough precision for detecting critical public health safety thresholds. The pH sensor accuracy of +/- 0.1 pH is sufficient to detect dangerous deviations from the safe range. The TDS sensor accuracy of +/- 10 ppm reliably identifies when concentrations exceed the 500 ppm WHO limit. The turbidity sensor accuracy of +/- 5% effectively detects filtration failures that could compromise water safety.

Table 8.2 Cost Comparison - Prototype vs Industrial Systems

Parameter	This Project	Industrial System
Initial Cost	Rs. 10.369	Rs. 1,00,000 - 2,50,000

Power Requirement	Solar + Battery (Off-grid)	Grid Power Required
Cloud Service Cost	Free (ThingSpeak)	Rs. 5,000 - 20,000/year
Installation Cost	Minimal (DIY)	Rs. 10,000 - 50,000
Maintenance	User Serviceable	Specialized Technician

The feasibility analysis demonstrates that this system offers a highly practical solution for rapid, mass deployment across rural drinking water tanks and urban residential complexes. The low cost enables installation at multiple points in the water distribution network, providing comprehensive coverage that expensive industrial systems cannot economically achieve. The autonomous operation reduces ongoing operational costs, while the instant alert capability ensures rapid response to water quality issues.

8.3 Future Scope

The current architecture provides a robust foundation for extensive future development and municipal scaling. Several enhancement directions have been identified that would further extend the system's capabilities and applicability.

8.3.1 LoRa Integration

For rural drinking water reservoirs where Wi-Fi hotspots are unavailable, future iterations will replace the Wi-Fi protocol with LoRa (SX1278) transceivers, enabling data transmission over distances of 2 to 5 kilometers to a central municipal gateway. LoRa's low power consumption and long-range capabilities make it ideal for remote monitoring applications.

8.3.2 Automated Chlorination Actuators

The system's edge-computing capabilities can be expanded from passive monitoring to active physical intervention. By connecting relay modules to the ESP32, the system could automatically trigger motorized dosing pumps to inject

precise amounts of liquid chlorine or pH buffers into the drinking water supply the exact moment the sensors detect biological vulnerability or an imbalance.

8.3.3 Machine Learning Predictive Analytics

By aggregating years of ThingSpeak data, machine learning models could be developed to predict seasonal contamination events before they occur, optimizing urban water treatment schedules. Pattern recognition algorithms could identify early warning signs of system degradation, enabling preventive maintenance before failures occur.

8.3.4 Additional Sensor Integration

Future versions could incorporate additional sensors for dissolved oxygen, conductivity, and specific contaminant detection (such as heavy metals or nitrates). The modular architecture supports sensor expansion without requiring fundamental system redesign.

8.3.5 Mobile Application Development

A dedicated mobile application could provide facility managers with real-time access to water quality data, historical trends, and alert management. The app could support multiple monitoring sites, providing a centralized dashboard for water utilities managing distributed infrastructure.

These future enhancements build upon the solid foundation established by this system, extending capabilities while maintaining the core principles of affordability, reliability, and autonomous operation.

Appendix: Source Code

The following Arduino C++ code implements the complete water quality monitoring system with edge computing, Deep Sleep power management, and cloud connectivity.

Main Firmware (WaterQualityMonitoring)

```
#include <WiFi.h>
#include <HTTPClient.h>
#include <OneWire.h>
#include <DallasTemperature.h>
#include <Wire.h>
#include <Adafruit_GFX.h>
#include <Adafruit_SSD1306.h>

// --- THE BROWNOUT BYPASS LIBRARIES ---
#include "soc/soc.h"
#include "soc/rtc_cntl_reg.h"

// --- WI-FI & THINGSPEAK CREDENTIALS ---
const char* ssid = "realme 9 5G Speed Edition";
const char* password = "12345678";
String apiKey = "9R36QCFMBABG8LGT";

// --- SENSOR PINS ---
#define ONE_WIRE_BUS 4    // Temp Sensor on Pin 4
#define PH_PIN 32
#define TDS_PIN 33
#define TURB_PIN 34

// --- OLED SETTINGS ---
#define SCREEN_WIDTH 128 // OLED display width, in pixels
#define SCREEN_HEIGHT 64 // OLED display height, in pixels
#define OLED_RESET     -1 // Reset pin # (or -1 if sharing Arduino rese
t pin)
Adafruit_SSD1306 display(SCREEN_WIDTH, SCREEN_HEIGHT, &Wire, OLED_RESET
);

// --- DEEP SLEEP SETTINGS ---
#define uS_TO_S_FACTOR 1000000ULL  // Conversion factor for micro secon
ds to seconds
#define TIME_TO_SLEEP  30          // Time ESP32 will go to sleep (in s
econds)

OneWire oneWire(ONE_WIRE_BUS);
DallasTemperature tempSensor(&oneWire);

void setup() {
  // *** CRITICAL BROWNOUT FIX ***
  WRITE_PERI_REG(RTC_CNTL_BROWN_OUT_REG, 0);

  Serial.begin(115200);
  Serial.println("\n\n=======================================");
  Serial.println("ESP32 WAKING UP FROM DEEP SLEEP");
```

```
Serial.println("========================================");

// Initialize OLED Display (Standard I2C address is 0x3C)
if(!display.begin(SSD1306_SWITCHCAPVCC, 0x3C)) {
  Serial.println(F("OLED allocation failed"));
} else {
  display.clearDisplay();
  display.setTextSize(1);
  display.setTextColor(WHITE);
  display.setCursor(0, 20);
  display.println("Booting up...");
  display.println("Connecting WiFi...");
  display.display();
}

tempSensor.begin();

// Connect to Wi-Fi with a timeout
WiFi.mode(WIFI_STA);
Serial.print("Connecting to Wi-Fi");
WiFi.begin(ssid, password);
int attempts = 0;
while (WiFi.status() != WL_CONNECTED && attempts < 20) {
  delay(500);
  Serial.print(".");
  attempts++;
}

if (WiFi.status() == WL_CONNECTED) {
  Serial.println("\nWi-Fi Connected!");

  // 1. Read Real Temperature
  tempSensor.requestTemperatures();
  float tempC = tempSensor.getTempCByIndex(0);
  float safeTemp = (tempC > -10.0) ? tempC : 25.0;

  // 2. Read & Calculate pH
  float phVolt = (analogRead(PH_PIN) * 3.3) / 4095.0;
  float phLevel = (13.42 * phVolt) - 8.07;
  if(phLevel < 0.0) phLevel = 0.0;
  if(phLevel > 14.0) phLevel = 14.0;

  // 3. Read & Calculate TDS (Compensated)
  float tdsVolt = (analogRead(TDS_PIN) * 3.3) / 4095.0;
  float compCoeff = 1.0 + 0.02 * (safeTemp - 25.0);
  float compVolt = tdsVolt / compCoeff;
  float tdsPPM = (133.42 * pow(compVolt, 3)
                  - 255.86 * pow(compVolt, 2)
                  + 857.39 * compVolt) * 0.5;
```

```
    if(tdsPPM < 0) tdsPPM = 0;

    // 4. Read & Calculate Turbidity
    float turbVolt = (analogRead(TURB_PIN) * 3.3) / 4095.0;

    // --- 3.3V CUSTOM CALIBRATION ---
    float cleanWaterVoltage = 1.57;
    float turbNTU = -(3000.0 / cleanWaterVoltage) * turbVolt + 3000.0;

    // --- THE AIR/CLEAN WATER LOCK ---
    if(turbVolt >= 1.45) {
      turbNTU = 0;
    }

    // Safety caps
    if(turbNTU < 0) turbNTU = 0;
    if(turbNTU > 3000) turbNTU = 3000;

    // ------------------------------------------------------
    // EDGE COMPUTING: WATER QUALITY INDEX (WQI)
    // ------------------------------------------------------
    float wqi = 100.0;
    wqi -= abs(phLevel - 7.0) * 8.0;
    wqi -= (tdsPPM * 0.05);
    wqi -= (turbNTU * 0.1);
    if (wqi < 0) wqi = 0;
    if (wqi > 100) wqi = 100;

    Serial.printf("Data -> Temp: %.2fC | pH: %.2f | TDS: %.0f ppm | Tur
b: %.0f NTU\n",
                  tempC, phLevel, tdsPPM, turbNTU);
    Serial.printf("WATER QUALITY INDEX (WQI): %.0f / 100\n", wqi);

    // ------------------------------------------------------
    // UPDATE OLED SCREEN
    // ------------------------------------------------------
    display.clearDisplay();
    display.setTextSize(1);
    display.setCursor(15, 0);
    display.print("SMART WATER BUOY");
    display.drawLine(0, 10, 128, 10, WHITE);
    display.setCursor(0, 15);
    display.printf("Temp: %.1f C\n", tempC);
    display.printf("pH  : %.1f\n", phLevel);
    display.printf("TDS : %.0f ppm\n", tdsPPM);
    display.printf("Turb: %.0f NTU\n", turbNTU);
    display.drawLine(0, 50, 128, 50, WHITE);
    display.setCursor(0, 54);
    display.printf("WQI SCORE: %.0f / 100", wqi);
```

```
    display.display();

    // 5. Upload to ThingSpeak
    Serial.println("Pushing to ThingSpeak...");
    HTTPClient http;
    String url = "http://api.thingspeak.com/update?api_key=" + apiKey +
                 "&field1=" + String(tempC) +
                 "&field2=" + String(phLevel) +
                 "&field3=" + String(tdsPPM) +
                 "&field4=" + String(turbNTU) +
                 "&field5=" + String(wqi);
    http.begin(url);
    int responseCode = http.GET();
    if (responseCode == 200) {
      Serial.println("Upload Success! (HTTP 200)");
    } else {
      Serial.println("ThingSpeak Error: " + String(responseCode));
    }
    http.end();

  } else {
    Serial.println("\nWi-Fi Failed.");
    display.clearDisplay();
    display.setCursor(0, 20);
    display.println("Wi-Fi Error!");
    display.display();
  }

  // ------------------------------------------------------
  // DEEP SLEEP COMMAND
  // ------------------------------------------------------
  Serial.println("Holding screen for 10 seconds...");
  delay(10000);
  display.clearDisplay();
  display.display();
  Serial.printf("Shutting down for %d seconds to save battery...\n", TI
ME_TO_SLEEP);
  Serial.println("========================================\n");

  esp_sleep_enable_timer_wakeup(TIME_TO_SLEEP * uS_TO_S_FACTOR);
  esp_deep_sleep_start();
}

void loop() {
  // Intentionally empty - all logic runs in setup()
}
```

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